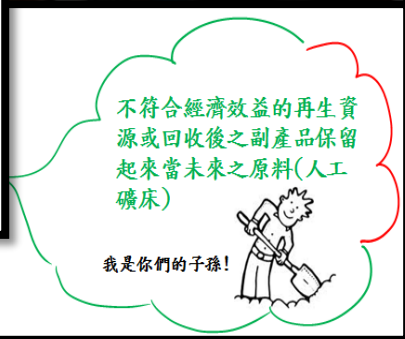
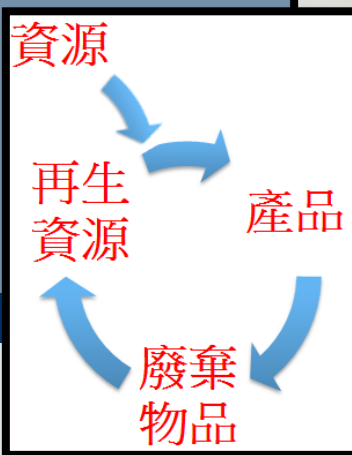


# 台日資源循環技術交流研討會

## 稀土元素回收技術與再生應用

報告者: 蘇昭瑾

日期: 2014/10/15



主辦單位：台灣資源再生協會、日本稀有金屬資源再生技術研究會  
 協辦單位：中華民國對外貿易發展協會、展盟展覽有限公司、台日產業合作推動辦公室(TJPO)





# 簡報綱要

台日資源循環技術交流研討會

1

稀有元素簡介與回收趨勢

2

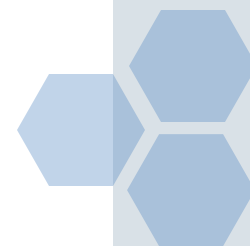
國外稀土元素回收技術

3

回收稀土元素應用實例

4

結論



稀有元素：在地殼中平均含量 $\leq 0.01\%$ 的化學元素

1

18

稀有輕金屬

稀有分散元素

難溶稀有金屬

稀有貴金屬

放射性稀有金屬

稀土金屬

lanthanoids

actinoids

Atomic Number

Symbol

Name

Standard Atomic Weight

Metals

Non-metals

Alkali Metals

Alkali Earth Metals

Transition Metals

Lanthanoids

Actinoids

Metalloids

Halogens

Noble Gases

hydrogen

1.007 94(7)

1

2

3

4

5

6

7

8

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13

14

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115

116

117

118

La-Lu

Ac-Lr

Rf

Db

Sg

Bh

Hs

Mt

Ds

Rg

La

Ce

Pr

Nd

Pm

Sm

Eu

Gd

Tb

Dy

Ho

Er

Tm

Yb

Lu

Ac

Th

Pa

U

Np

Pu

Am

Cm

Bk

Cf

Es

Fm

Md

No

Lr

potassium

calcium

scandium

titanium

vanadium

chromium

manganese

iron

cobalt

nickel

copper

zinc

gallium

germanium

arsenic

selenium

bromine

krypton

rubidium

strontium

yttrium

zirconium

niobium

molybdenum

technetium

ruthenium

rhodium

palladium

silver

cadmium

indium

tin

antimony

tellurium

iodine

xenon

caesium

barium

lanthanoids

hafnium

tantalum

tungsten

rhenium

osmium

iridium

platinum

gold

mercury

thallium

lead

bismuth

polonium

astatine

radon

francium

radium

actinoids

rutherfordium

dubnium

seaborgium

bohrium

hassium

meitnerium

darmstadtium

roentgenium

lanthanum

cerium

praseodymium

neodymium

promethium

samarium

europium

gadolinium

terbium

dysprosium

holmium

erbium

thulium

ytterbium

lutetium

actinium

thorium

protactinium

uranium

neptunium

plutonium

americium

curium

berkelium

californium

einsteinium

fermium

mendelevium

nobelium

lawrencium

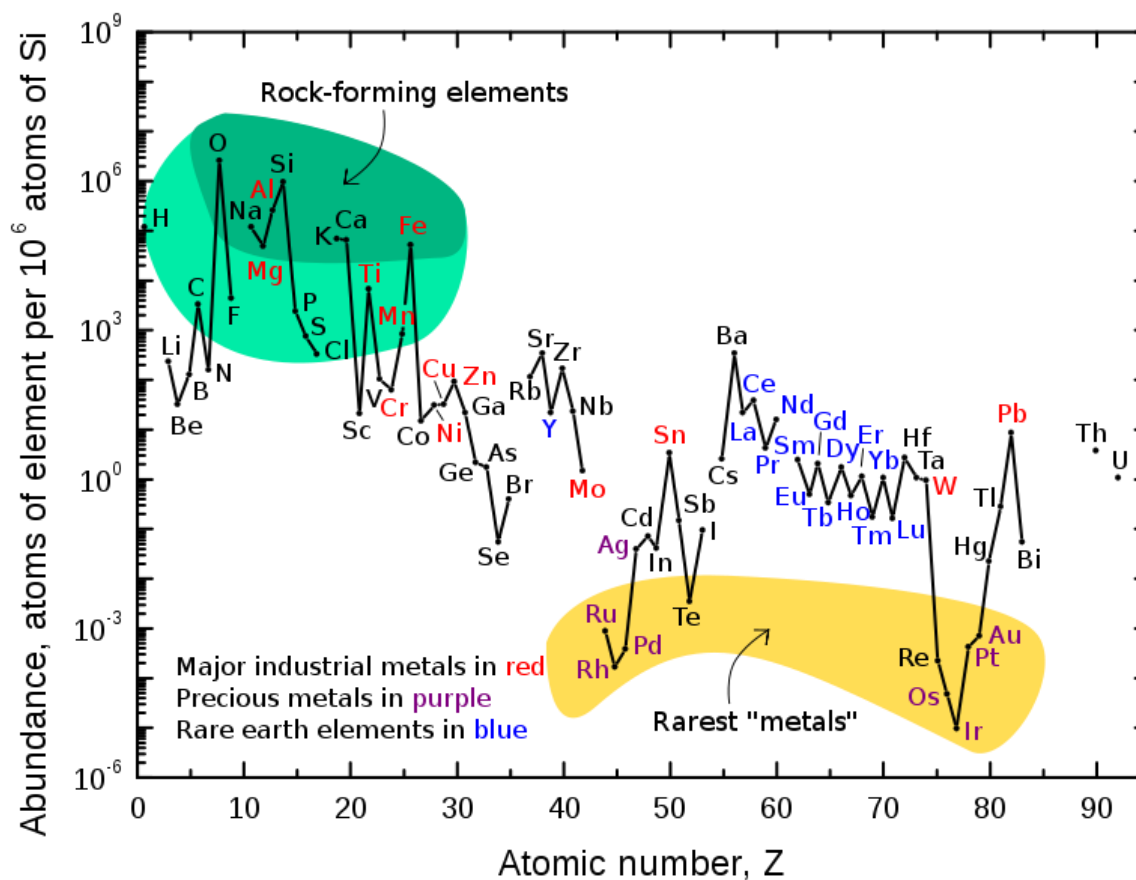
©2009 HowStuffWorks Source: International Union of Pure and Applied Chemistry



# 稀土元素與稀有元素

台日資源循環技術交流研討會

## 稀土元素並不一定稀有



[http://en.wikipedia.org/wiki/File:Elemental\\_abundances.svg](http://en.wikipedia.org/wiki/File:Elemental_abundances.svg)

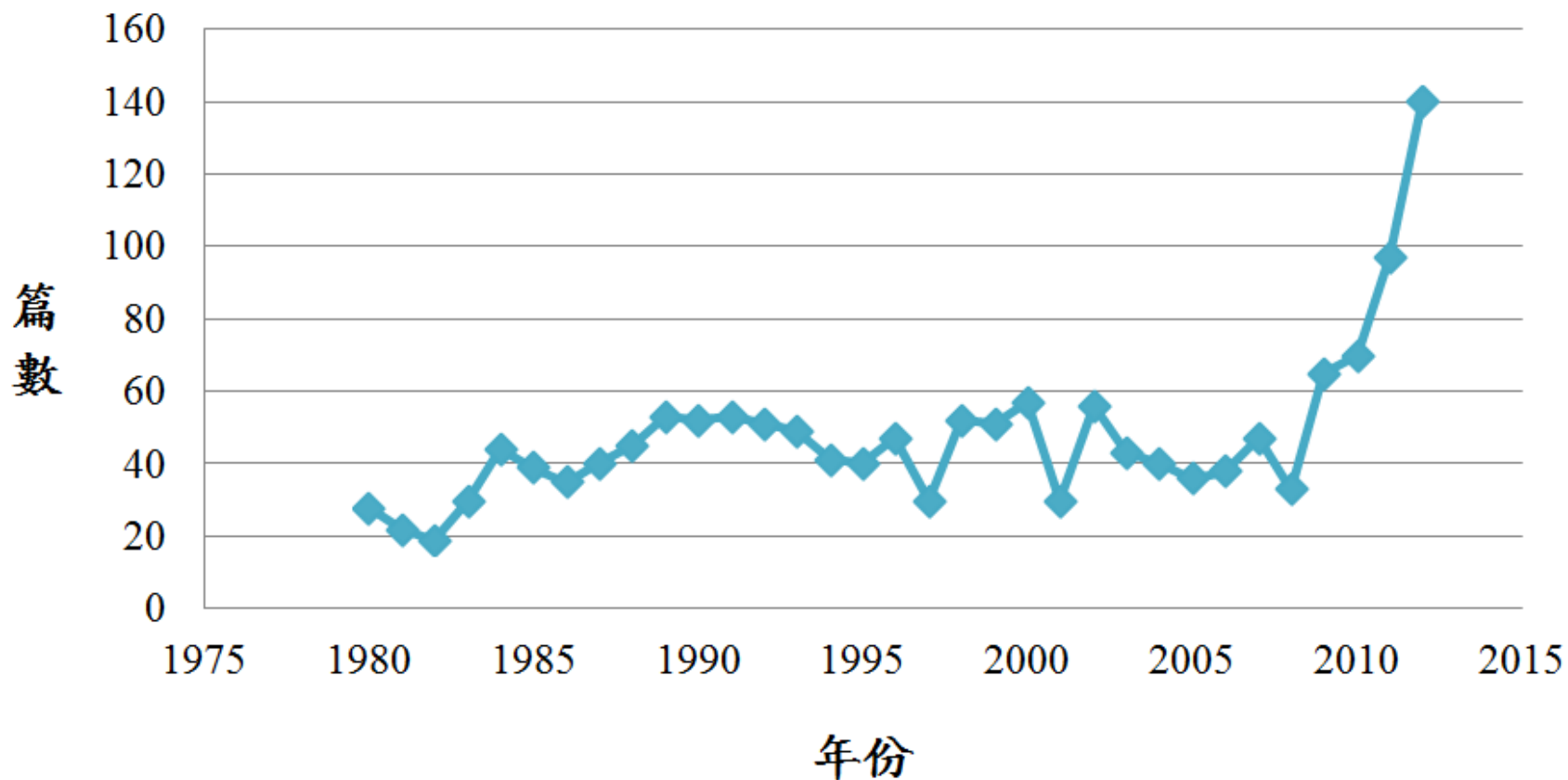


# 稀土金屬回收發展圖

2014年資源廢棄物能資源化技術研討會

## ❖ 指數型的成長

2010年中國大陸限制稀土出口

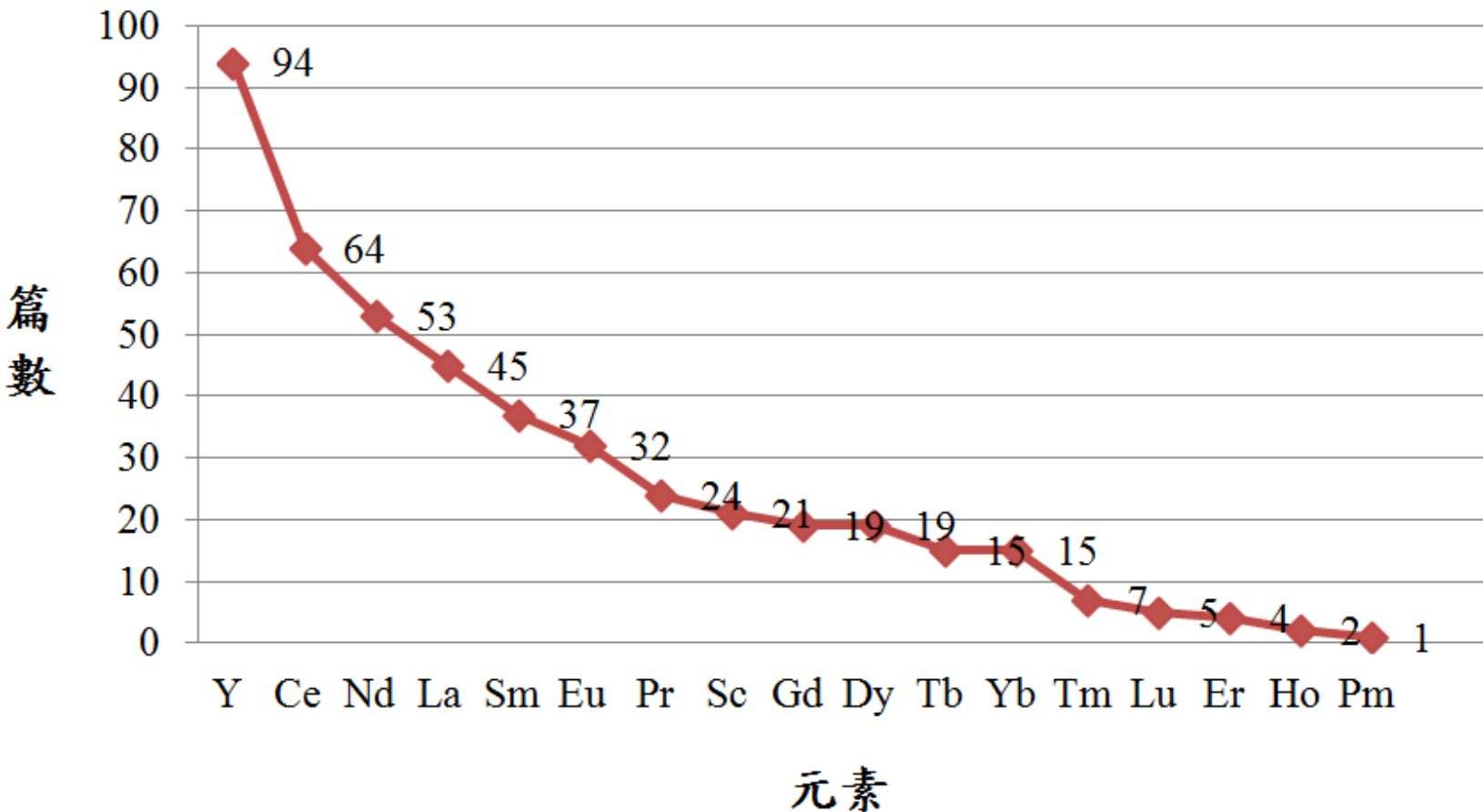




# 稀土金屬個別元素回收比較圖

台日資源循環技術交流研討會

- ❖ 鈮、鈰、釹、鐳、鎳、鎳以及釷等稀土金屬回收的技術是較廣為探討的研究區塊

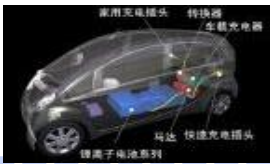




# 稀土金屬個別元素回收比較圖

台日資源循環技術交流研討會

- ❖ 鈹、鈾、釷、釷、鐳、鎢以及鈦等稀土金屬回收的技術是較廣為探討的研究區塊

| 照明                                                                                                                                                                  | 環保                                                                                                                                                                   | 建築                                                                                             | 國防                                                                                                                                                                                                                                                                        |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|   |   |             |                                                                                                     |
| LED燈<br>(Y、Eu、La...等稀土<br>螢光粉)                                                                                                                                      | 太陽能板<br>(Ga) 風力發電機<br>(Nd)                                                                                                                                           | 高速鋼(Co)、<br>無縫鋼管(Mn、V)                                                                         | 坦克的導航系統<br>(Sm、Co、Nd) 導彈雷達系統<br>(Sm、Co、Nd)                                                                                                                                                                                                                                |
| 醫療                                                                                                                                                                  | 交通運輸                                                                                                                                                                 | 資通訊                                                                                            | 娛樂                                                                                                                                                                                                                                                                        |
| MRI(Nd)、<br>造影劑(Gd)<br>                                                           | 磁浮列車<br>(Nd、Sm、<br>Pr、Dy)<br>                                                      | 半導體(Ga)<br> | 光纖(Eu、Y)<br>   |
| 人工牙齒<br>(Co、Cr、<br>Ti 等)<br>                                                     | 電動車<br>馬達<br>(Nd)<br>                                                             | 電池(Li)<br> | 手機 (Nd、<br>Ta、Ni)、<br>觸控面板<br>(In) 數位相機鏡頭(Ta)<br>                                                                                                                                    |



# 稀土金屬個別元素回收比較圖

台日資源循環技術交流研討會

釹

日立 風熨斗滾筒洗衣機(SFBD3300T)



- 採用Neo56DD直驅式馬達(56顆釹鐵硼磁石)
- 脫水轉速最高1,600rpm
- 容量12公斤

售價：72,900元

一般

某國產品牌滾筒式洗衣機(TAW-A120T)



- 直驅變頻靜音馬達
- 六段脫水轉速，最高1,200rpm
- 容量12公斤

售價：42,900元

1.7倍

Dyson吸塵器(DC26 Allergy)



- 輕巧可攜3.22公斤
- 寬250、深450、高300(mm)
- 免集塵袋，容量0.68公升
- 數位(釹磁鐵)馬達(永不減弱)

售價：25,500元

某國產品牌 家用吸塵器(SC-128)



- 重量5.5公斤
- 寬320、深245、高210(mm)
- 需集塵袋

售價：1,390元

18.5倍

日立超值變頻冷暖氣機(RAS-36RU)



- DC變頻冷氣(永磁無刷馬達)
- 6-8坪
- EER：3.85(4.00~3.50)kW/kW
- 3.31(3.44~3.01)kcal/hW

售價：49,900元

某國產品牌 窗型冷氣(TAW-A20)



- 4-6坪
- EER2.33

售價：12,500元

4.0倍

Dyson air multiplier AM01風扇



- 無扇葉電風扇
- 使用數位(釹磁鐵)馬達
- 360度迴轉送風
- 每秒可吹出405公升的涼風

售價：26,999元

某國產品牌 渦流循環扇(SF-1058)



- 強、中、弱三段風速
- 100%台灣研發
- 整台於台灣生產製造

售價：1,400元

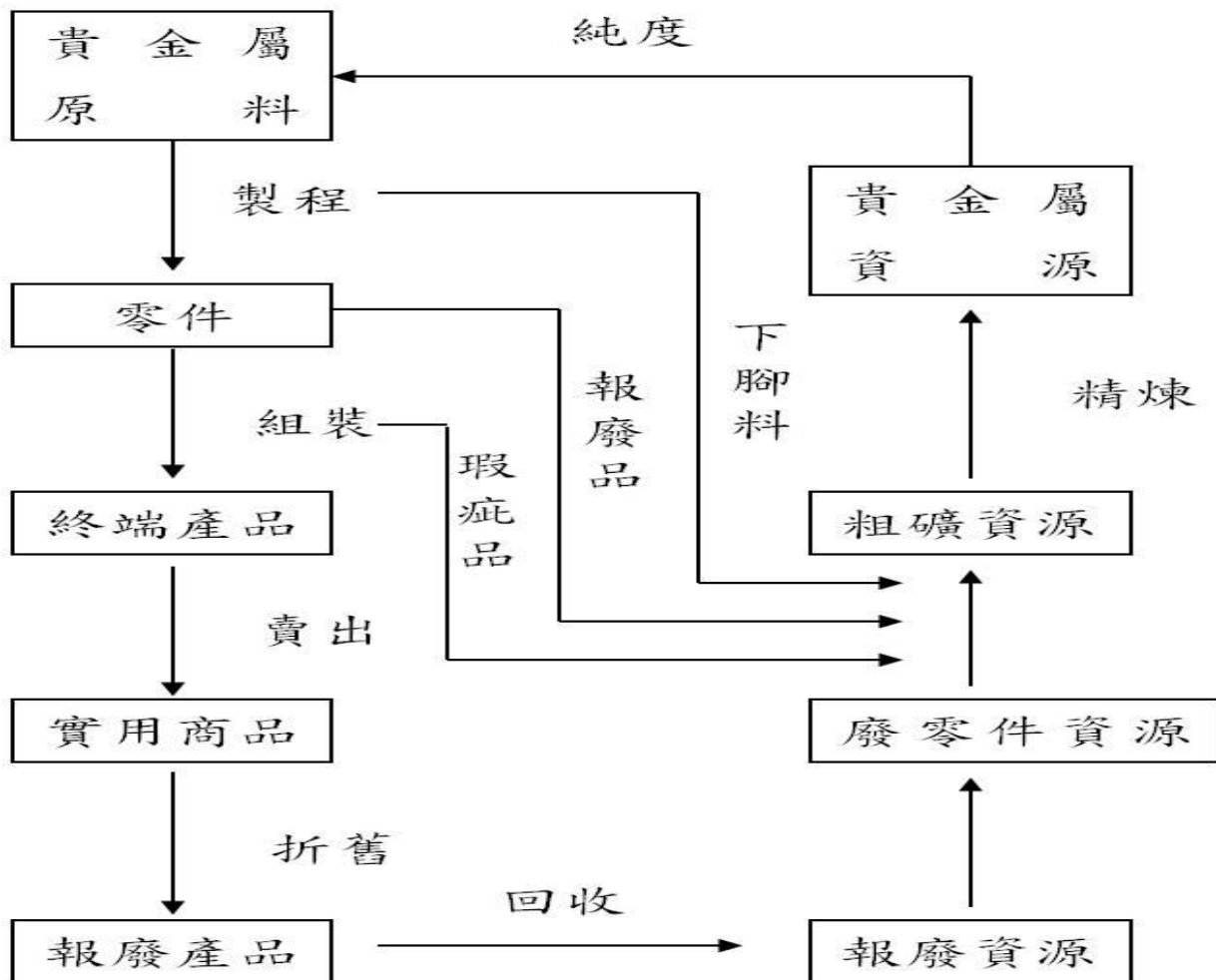
19.3倍

資料來源：資策會，稀土金屬與金屬之國際資源競爭與台灣因應對策分析簡報資料，100年12月



# 稀貴金屬回收產業鏈

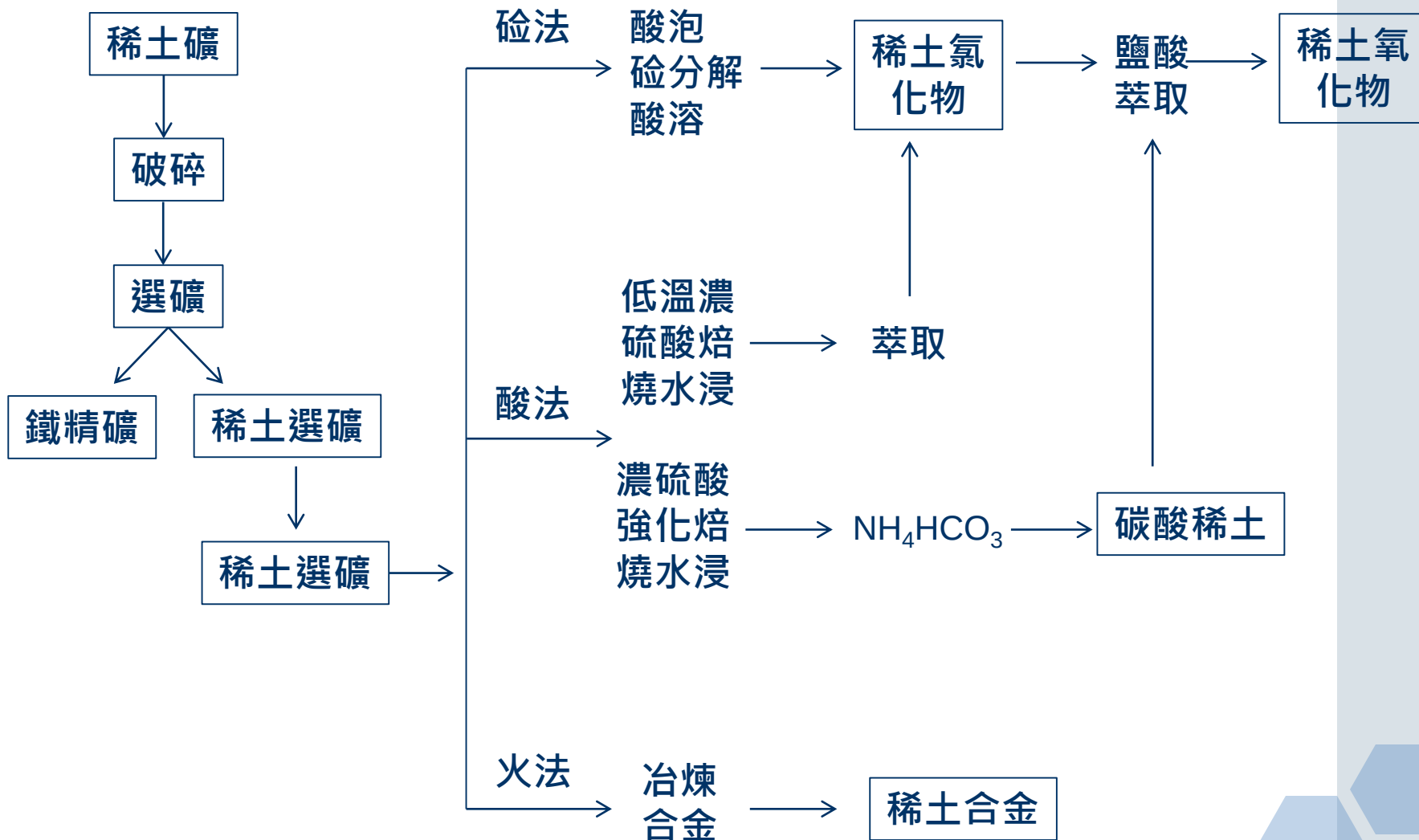
台日資源循環技術交流研討會





# 稀土物質生產製程

台日資源循環技術交流研討會





# 稀土元素回收技術

台日資源循環技術交流研討會

| 技術名稱         | 最終產品                                                                   | 回收率        | 國家  | 回收物品 | 優點                          | 缺點                                                                        | 文獻 |
|--------------|------------------------------------------------------------------------|------------|-----|------|-----------------------------|---------------------------------------------------------------------------|----|
| 酸浸法          | $Y_2(C_2O_4)_3$                                                        | 85%        | 義大利 | 螢光燈管 | 1. 方法簡易。<br>2. 硫酸可以降低鈣的溶出量。 | 1. 其他燈管成份未知。<br>2. 改成硝酸時，會產生有毒氣體 ( NO、NO <sub>2</sub> 、N <sub>2</sub> O ) | 1  |
| 酸浸法          | $Y_2O_3$<br>$Eu_2O_3$                                                  | 95%        | 大陸  | 螢光燈管 | 1. 回收率高。<br>2. 氧化釔的純度高。     | 其他燈管成份未知                                                                  | 2  |
| 酸浸法          | $Y_2(C_2O_4)_3$<br>$Eu_2(C_2O_4)_3$                                    | 98%<br>96% | 中國  | 螢光粉  | 1. 製成簡單。<br>2. 回收率高。        | 釔和鎔需要再進一步分離                                                               | 3  |
| 酸浸法<br>溶劑萃取法 | $La^{3+}$ 、 $Ce^{3+}$<br>$Pr^{3+}$ 、 $Eu^{3+}$<br>$Y^{3+}$ 、 $Tb^{3+}$ | >90%<br>%  | 日本  | 螢光粉  | 回收效率高                       | 需要投入較多能源                                                                  | 4  |
| 酸浸法<br>溶劑萃取法 | Y、Eu                                                                   | 98%<br>96% | 埃及  | 螢光粉  | 1. 純度高<br>2. 可工業化           | 製程複雜                                                                      | 5  |



# 稀土元素回收技術

台日資源循環技術交流研討會

| 技術名稱         | 最終產品                                                                           | 回收率        | 國家  | 回收物品    | 優點                        | 缺點                     | 文獻 |
|--------------|--------------------------------------------------------------------------------|------------|-----|---------|---------------------------|------------------------|----|
| 酸浸法          | $\text{La}_2\text{SO}_4$<br>$\text{Ce}(\text{SO}_4)_2$                         | 99%        | 義大利 | 鎳氫電池    | 回收率高                      | 鐳和鈾需再進一步分離             | 6  |
| 酸浸法          | $\text{LaCl}_3$<br>$\text{NdCl}_3$                                             | 98%        | 埃及  | 鎳氫電池    | 1. 環保、無有害附產品<br>2. 吸附劑可再生 | 製程較複雜                  | 7  |
| 酸浸法<br>溶劑萃取法 | $\text{La}^{3+}$                                                               | 99%        | 巴西  | 鎳氫電池    | 減少廢水產生                    | 需要額外純化                 | 8  |
| 酸浸法          | $\text{Nd}_2(\text{C}_2\text{O}_4)_3$<br>$\text{Dy}_2(\text{C}_2\text{O}_4)_3$ | 70%<br>51% | 日本  | 釹鐵硼磁性廢料 | 製程簡單                      | 去除鐵及草酸沉澱這兩步驟會損失釹和鐳的回收量 | 9  |
| 酸浸法          | $\text{Sc}_2\text{O}_3$                                                        | 60%        | 中國  | 礦       | 純度高                       | 回收率低                   | 10 |
| 溶劑萃取法        | La、Nd、Pr                                                                       | 96%        | 印度  | 礦       | 1. 回收率高<br>2. 純度高         | 製程較繁複                  | 11 |



## 參考文獻

台日資源循環技術交流研討會

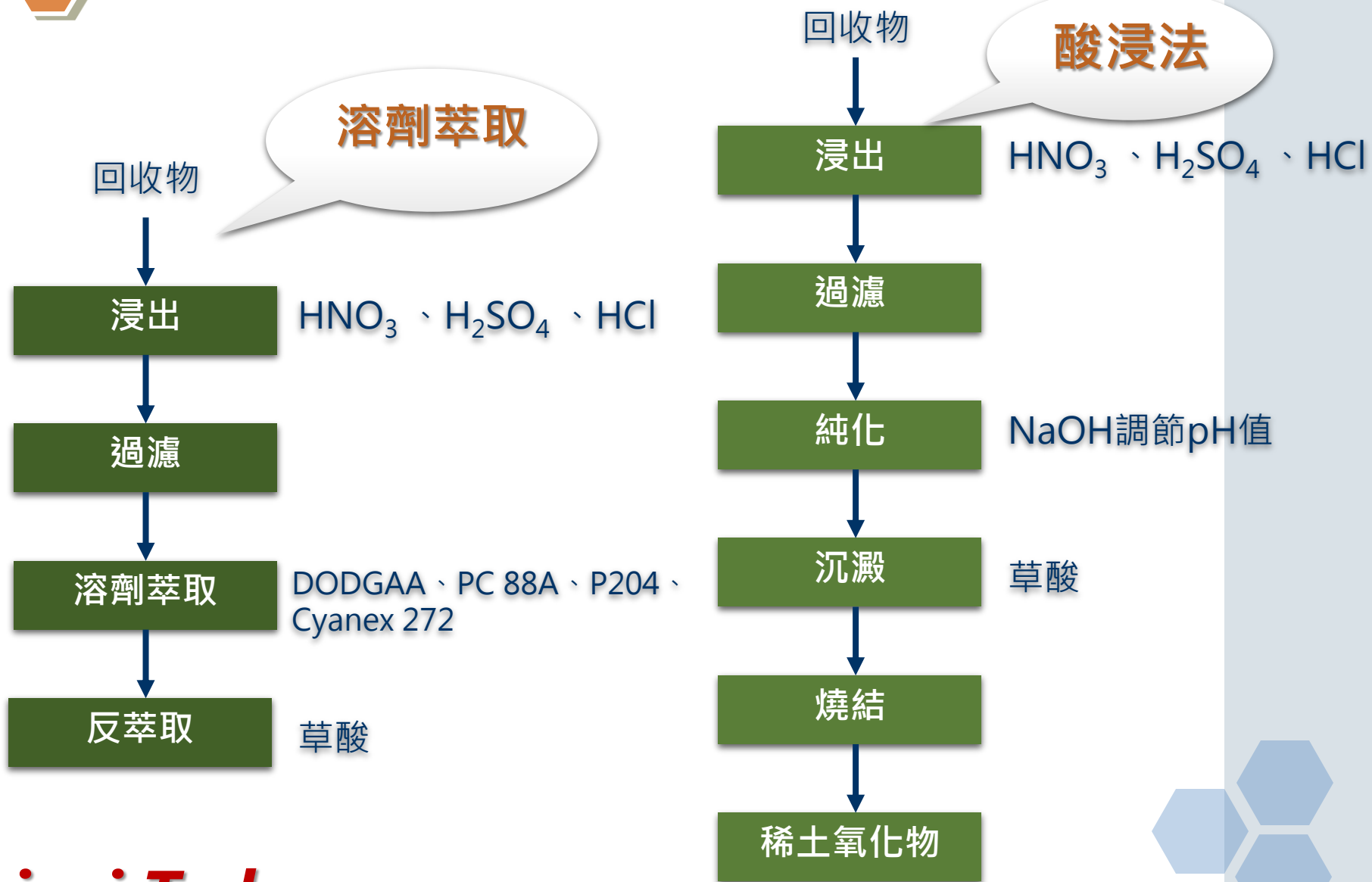
- ❖ 1. Waste Management 31 (2011) 2559–2568.
- ❖ 2. 5th International Conference on Bioinformatics and Biomedical Engineering, iCBBE 2011, art. no. 5781464
- ❖ 3. Applied Mechanics and Materials 295-298 (2013) 1840-1845.
- ❖ 4. Journal of Hazardous Materials 254– 255 (2013) 79-88.
- ❖ 5. Waste Management 28 (2008) 318–325.
- ❖ 6. Journal of Power Sources 211 (2012) 184-191.
- ❖ 7. International Journal of Mineral Processing 121 (2013) 31–38.
- ❖ 8. Hydrometallurgy 133 (2013) 37-43.
- ❖ 9. J Mater Cycles Waste Manag (2012) 1-8.
- ❖ 10. Chinese Journal of Rare Metals 36 (2012) 501-506.
- ❖ 11. Separation Science and Technology 47 (2012) 1644-1650.





# 稀土回收技術

台日資源循環技術交流研討會



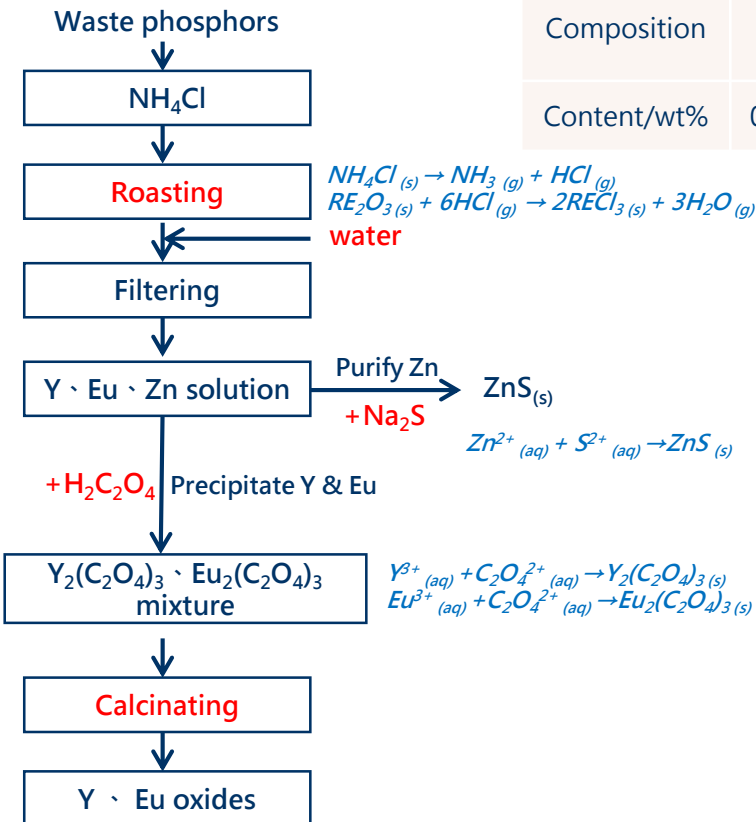


# Y and Eu from waste phosphors (CRT)

台日資源循環技術交流研討會

Main elements in phosphors waste tested by XRF :

| Composition | Al    | Y      | S      | Eu    | Ca    | Fe    | Zn     | Other composites |
|-------------|-------|--------|--------|-------|-------|-------|--------|------------------|
| Content/wt% | 0.984 | 25.515 | 21.335 | 1.736 | 0.866 | 1.462 | 47.590 | 0.512            |



## Remark

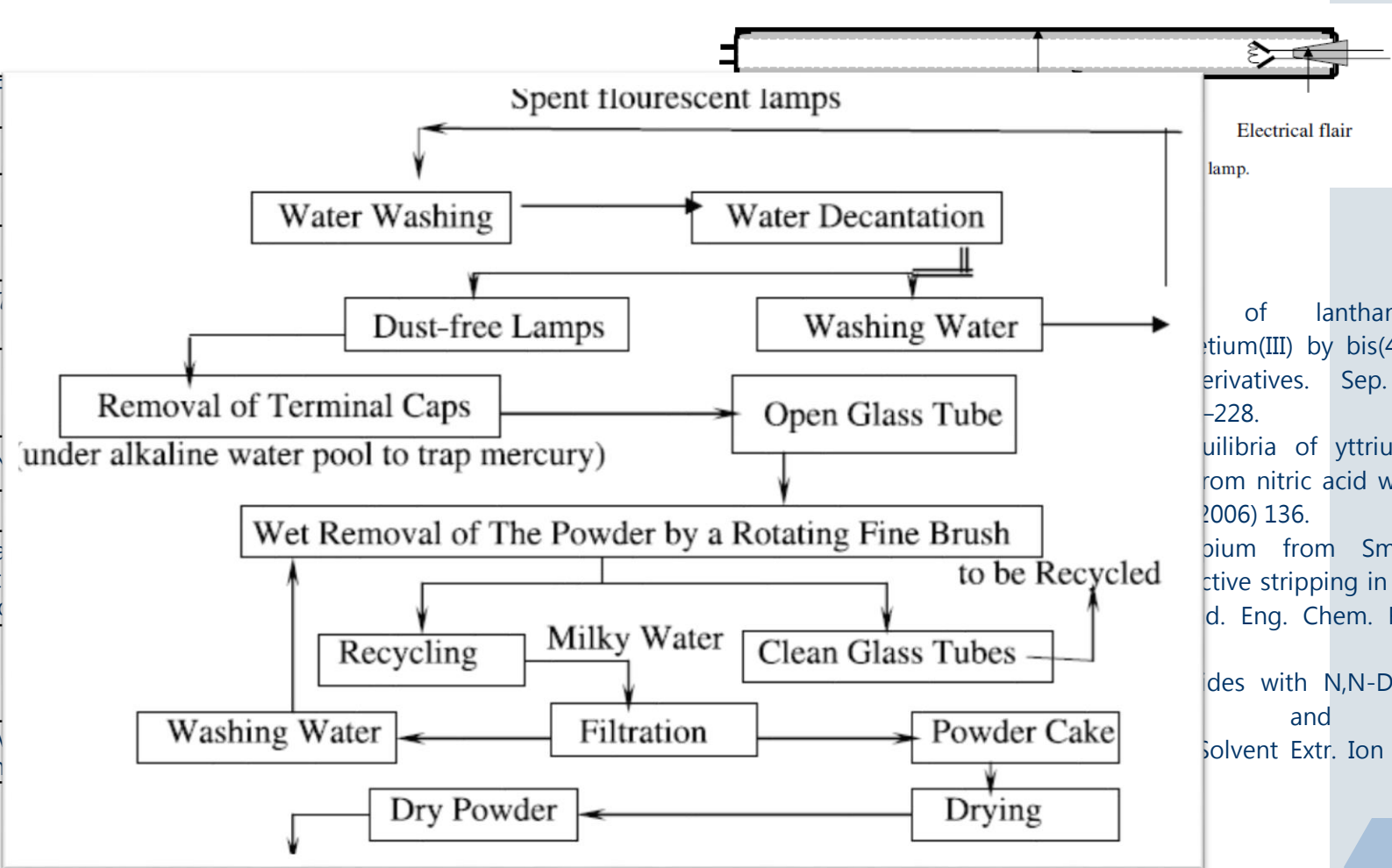
Yttrium (Y) and Europium (Eu) were recovered from waste phosphors of TV screens via several processes, including (1) roasting waste phosphors with ammonia chloride, (2) dissolving yttrium chloride and europium chloride with water from roasted waste phosphors, (3) purification with sodium sulfide, precipitating in the presence oxalic acid and (4) calcining at high temperature to form yttrium europium oxides which produce the red light when the excitation wavelength was 242 nm.

*Applied Mechanics and Materials 295-298 (2013) 1840-1845.*



# Y and Eu from spent fluorescent lamps

台日資源循環技術交流研討會



of lanthanum(III),  
tium(III) by bis(4acyl-5-  
erivatives. Sep. Purif.  
-228.

uilbria of yttrium and  
rom nitric acid with PC-  
2006) 136.

mium from Sm/Eu/Gd  
ctive stripping in solvent  
d. Eng. Chem. Rev. 34

des with N,N-Dimethyl  
and -3,6-  
Solvent Extr. Ion Exc. 22

*Waste Management 28 (2008) 318–325.*

# Economic study from 1000 kg spent lamps

台日資源循環技術交流研討會

Cost of recovery of europium and yttrium from 1000 kg spent lamps.  
(about 5000 spent lamps with different types)

| Item                               | Unit           | Quantity | Unit price (\$) | Subtotal (\$) | Total (\$) |
|------------------------------------|----------------|----------|-----------------|---------------|------------|
| <i>Running cost and utilities:</i> |                |          |                 |               |            |
| Spent lamps                        | Nr.            | 5000     | 0.01            | 50,000        |            |
| Transportation                     |                |          |                 | 12,000        |            |
| Storage                            |                |          |                 | 08,000        | 70,000     |
| <i>Chemicals (consumable)</i>      |                |          |                 |               |            |
| Sulfuric acid, 10 M                | kg             | 8.1      | 0.44            | 03,564        |            |
| Nitric acid, 6 M                   | kg             | 24.5     | 0.82            | 20,090        |            |
| Hydrochloric acid, 6 M             | kg             | 5        | 0.4             | 02,000        |            |
| Potassium thiocyanate              | kg             | 6.48     | 3.85            | 24,948        |            |
| Hydrogen gas                       | m <sup>3</sup> | 25*      | 1.90            | 22,750        |            |
| Water                              | m <sup>3</sup> | 1        | 0.11            | 00,110        | 73,462     |
| <i>Chemicals (recyclable)</i>      |                |          |                 |               |            |
| Organic solvent**<br>(3-4 % loss)  | kg             | 1.75     | 22.5            | 39,750        |            |
| Ethyl alcohol**<br>(5-7% loss)     | kg             | 1.5      | 4.75            | 06,125        | 45,875     |
| Energy, natural gas                | m3             | 120      | 0.08            | 09,600        |            |
| Power                              | k.w h          | 240      | 0.05            | 12,000        | 21,600     |
| Others                             |                |          |                 | 05,000        | 05,000     |
| Grand total                        |                |          |                 |               | 215,937    |
| <i>Weight of products:</i>         |                |          |                 |               |            |
| 8.6 kg europium                    | kg             | 6.54     |                 |               |            |
| 6.5 kg yttrium                     | kg             | 8.25     |                 |               |            |
| Average cost of recycling/kg       |                |          |                 |               | 15,310     |

\*\* Distilled commercial grade.

Average cost of recycling/kg = 15,310



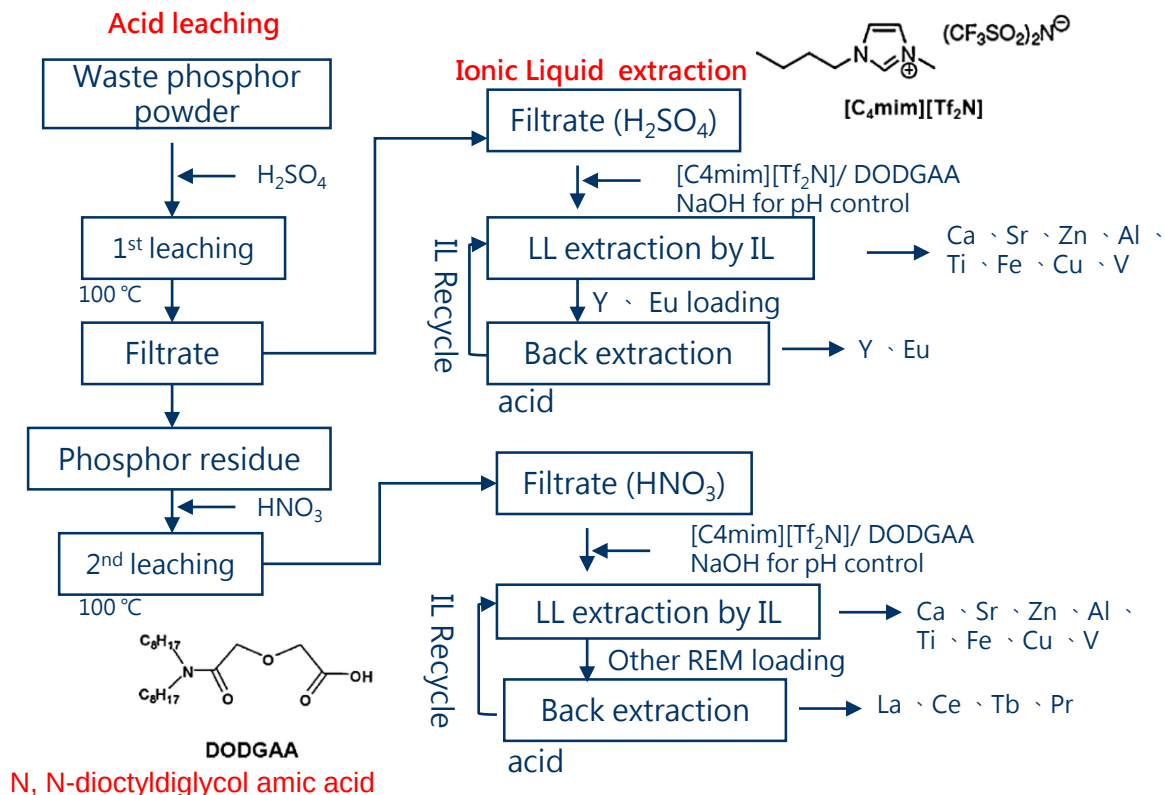
# REM from waste phosphors by ionic liquid

台日資源循環技術交流研討會

Imidazolium IL Liquid-liquid extraction is one of the most promising methods for the separation of rare earth metals. Metals contained in the scrap are leached by acid solution followed by the extraction and recovery of target metal ions from the leach solutions.

ionic liquids:

1. a "green" solvent
2. Flame resistance
3. Low vapor pressure
4. Readily tunable



High selectivity compared to the conventional PC-88A system. Y and Eu were dissolved readily into the acid solution, however, the leaching of other metals need large amount of energy.

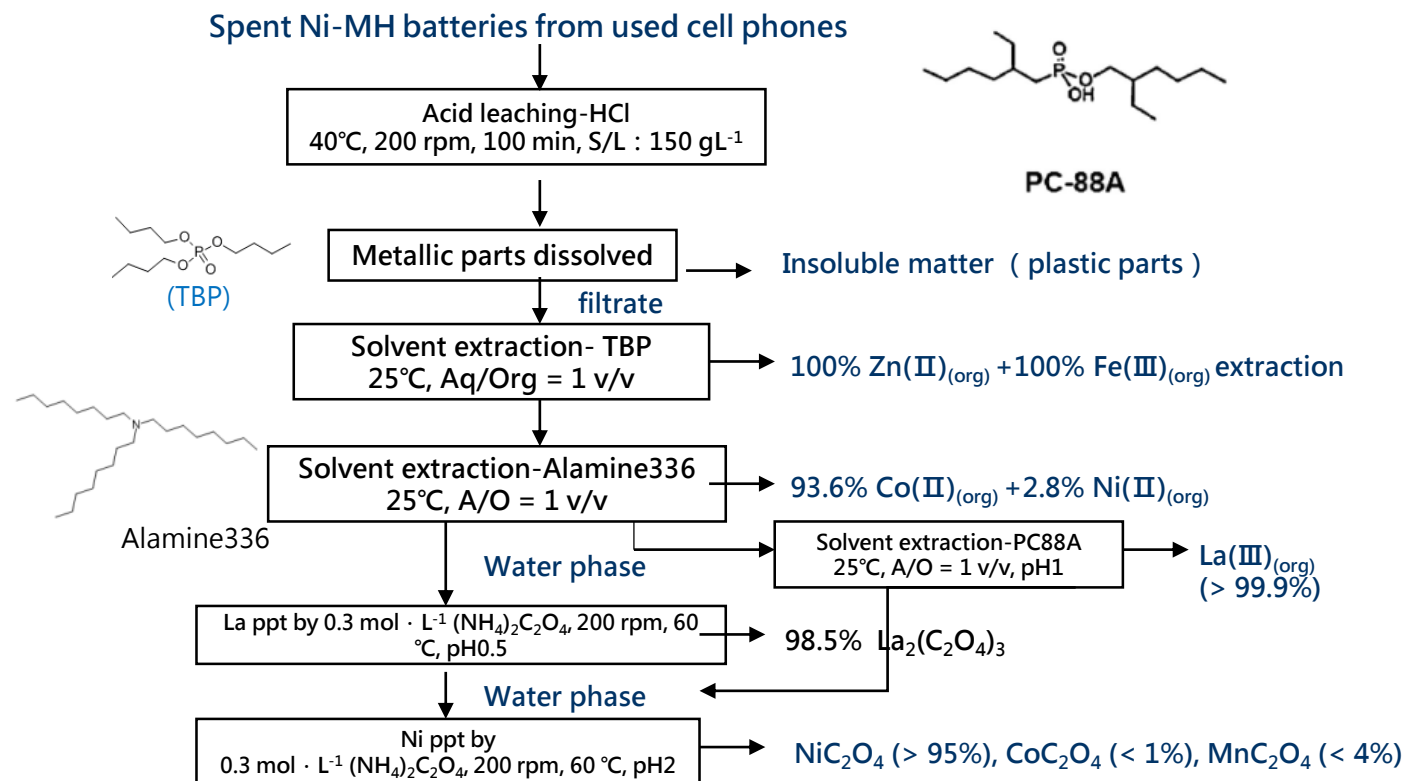
*Journal of Hazardous Materials 254–255 (2013) 79–88.*



# Lanthanides from spent Ni-MH batteries

台日資源循環技術交流研討會

MH: metal hydride

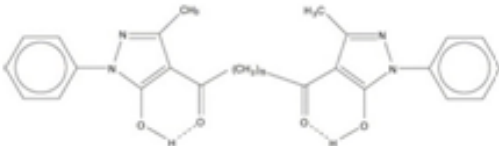
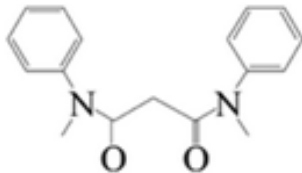
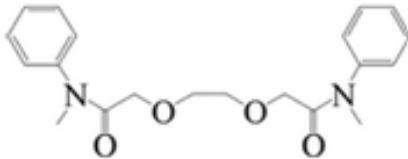
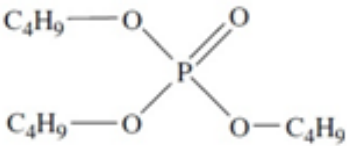


## Remark :

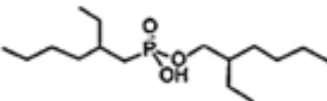
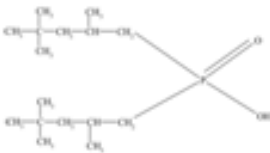
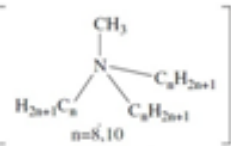
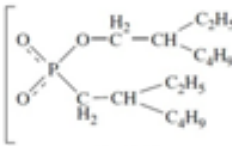
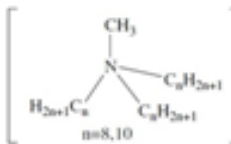
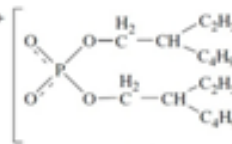
- Tertiary amines is a good extractants for Co<sup>2+</sup> in high HCl concentration.
- La were extracted (>99.9%) in 2 stage process with PC 88A.

*Hydrometallurgy 133 (2013) 37-43.*



| NO. | structure                                                                                                                                                                       | Recovery of metals                                       | References                                                            |
|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|-----------------------------------------------------------------------|
| 1   |  <p>bis(4-acyl-5-hydroxypyrazoles) HL-n-LH</p>                                                 | lanthanum (III),<br>europium (III) and<br>lutetium (III) | Separation and<br>Purification<br>Technology 50 (2006)<br>220-228     |
| 2   |  <p>bidentate <i>N,N'</i>-dimethyl-<i>N,N'</i>-diphenyl-malonamide (MA)</p>                    | Lanthanides (III)                                        | Solvent Extraction<br>and Ion Exchange<br>Volume 22 (2004)<br>135-145 |
|     |  <p>tetradentate <i>N,N'</i>-dimethyl-<i>N,N'</i>-diphenyl-3,6-dioxaoctanediamide (DOODA)</p> |                                                          |                                                                       |
| 3   |  <p>tri-n-butylphosphate (TBP)</p>                                                           | Yttrium (III),<br>Europium (III)                         | J. of Supercritical<br>Fluids 33 (2005) 235-<br>241                   |



| NO. | structure                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Recovery of metals                                                                            | References                                                      |
|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|-----------------------------------------------------------------|
| 4   |  <p>2-ethylhexylphosphonic acid, mono-2-ethylhexyl ester (PC-88A)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Yttrium (III),<br>Europium (III)                                                              | Materials<br>Transactions 47<br>(2006) 136-142                  |
| 5   |  <p>bis(2,4,4-trimethylpentyl)phosphinic acid (Cyanax 272)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Scandium (III),<br>Yttrium (III),<br>Lanthanum (III),<br>Gadolinium (III),<br>Ytterbium (III) | Separation and<br>Purification<br>Technology 50<br>(2006) 30-34 |
| 6   | <div style="display: flex; justify-content: space-around; align-items: center;"> <div style="text-align: center;">  <p>[A336]<sup>+</sup></p> </div> <div style="text-align: center;">  <p>[P507]<sup>-</sup></p> </div> </div> <p>[N-Methyl-N,N-di(octyl)octan-1-ammonium chloride, A336][ 2-ethylhexylphosphoric acid mono (2-ethylhexyl) ester, P507]</p> <div style="display: flex; justify-content: space-around; align-items: center;"> <div style="text-align: center;">  <p>[A336]<sup>+</sup></p> </div> <div style="text-align: center;">  <p>[P204]<sup>-</sup></p> </div> </div> <p>[N-Methyl-N,N-di(octyl)octan-1-ammonium chloride, A336][ di(2-ethylhexyl)phosphoric acid, P204]</p> | RE (III)                                                                                      | J Chem Technol<br>Biotechnol 87 (2012)<br>198-205               |



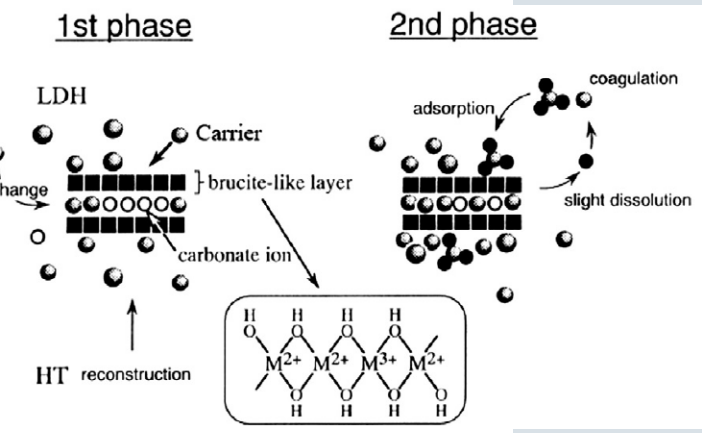
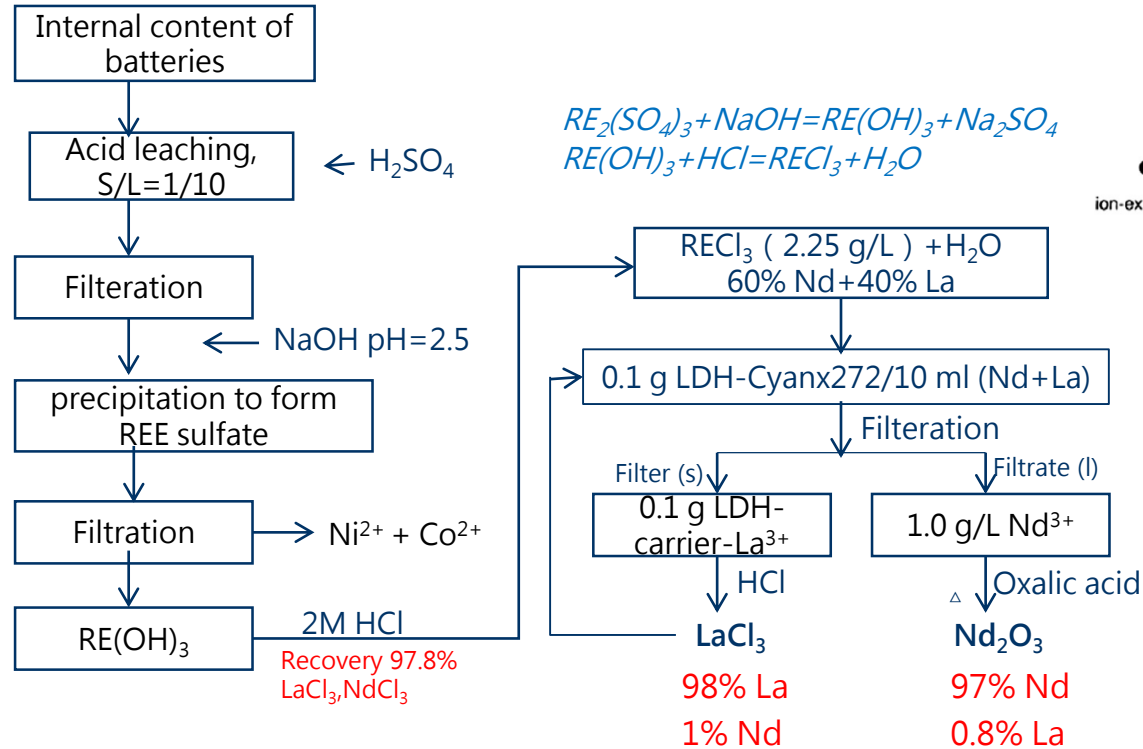


| NO. | structure                                                                                                                                                                                                                                                                                                                          | Recovery of metals | References                                        |
|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|---------------------------------------------------|
| 6   | $  \begin{array}{c}  \text{C}_4\text{H}_9-\text{O} \\  \diagup \quad \diagdown \\  \text{P}=\text{O} \\  \diagdown \quad \diagup \\  \text{C}_4\text{H}_9-\text{O} \quad \text{O}-\text{C}_4\text{H}_9  \end{array}  $ <p>tri-n-butylphosphate (TBP)</p>                                                                           | RE (III)           | J Chem Technol<br>Biotechnol 87 (2012)<br>198-205 |
|     | $  \begin{array}{c}  \text{CH}_3 \\    \\  \text{CH}_3(\text{CH}_2)_5-\text{CH}-\text{O} \\  \diagup \quad \diagdown \\  \text{P}=\text{O} \\  \diagdown \quad \diagup \\  \text{CH}_3(\text{CH}_2)_5-\text{CH}-\text{O} \quad \text{CH}_3 \\    \\  \text{CH}_3  \end{array}  $ <p>di-(1-methylheptyl)methyl phosphate (P350)</p> |                    |                                                   |
|     | $  \begin{array}{c}  \text{CH}_3(\text{CH}_2)_7 \\  \diagdown \\  \text{CH}_3(\text{CH}_2)_7-\text{P}=\text{O} \\  \diagup \\  \text{CH}_3(\text{CH}_2)_5  \end{array}  $ <p>mixture of straight chain alkylated phosphine oxides<br/>(Cyanex923)</p>                                                                              |                    |                                                   |





# La & Nd from spent Ni-MH batteries



• The adsorption capacities of the adsorbent for  $\text{La}^{3+}$  and  $\text{Nd}^{3+}$  were 481 and 192 mg/g.

※吸附劑：MgFe-LDH-Cyanx272

Layered double hydroxides (LDHs)

- The Mag-Fe-LDH-A can be recovered by using 0.01 M  $\text{HCl}$ . And the recovered adsorbent was reused to uptake REEs.
- Ecomic and efficient, LDH used several times.

7. International Journal of Mineral Processing 121 (2013) 31–38.



# 「重金屬回收劑」

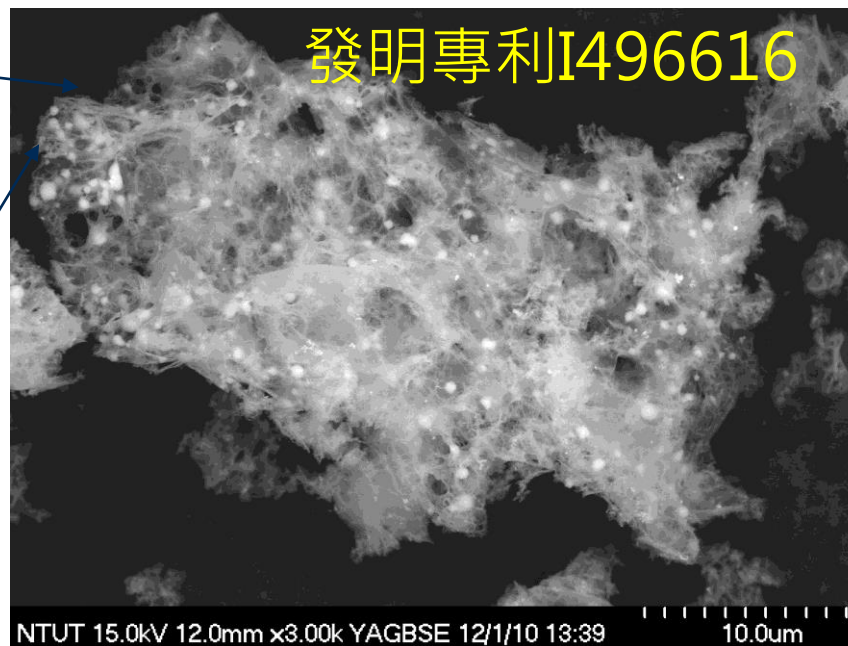
台日資源循環技術交流研討會

## CATALYST FOR HEAVY METAL RECOVERY AND METHODS OF PRODUCTION AND USE THE SAME

重金屬回收劑包含一氧化物粉末，可以是赤銅鐵礦型、尖晶石型或其混合物

具有棉花糖形狀的  
光觸媒

照光下將溶液中的  
重金屬還原  
形成大顆粒  
吸附在光觸媒上



表面積至少  $>10\text{m}^2/\text{g}$

$\text{AxByOz}$ 代表，A:銀(Ag)、銅(Cu)、鈀(Pd)、鉑(Pt)，

B:鋁(Al)、鈦(Sc)、鉻(Cr)、釔(Y)、鐵(Fe)、銦(In)、鎳(Ga)、  
鈷(Co)、錳(Mn)、銦(Rh)、鎳(Ni)、鐳(La)、釹(Nd)、釷(Sm)、鎔(Eu)、鈦(Ti)

(邱德威、楊重光)、蘇昭瑾



# 「奈米金屬/複合氧化物觸媒及其合成方法」

台日資源循環技術交流研討會

中華民國專利，發明第I430837號

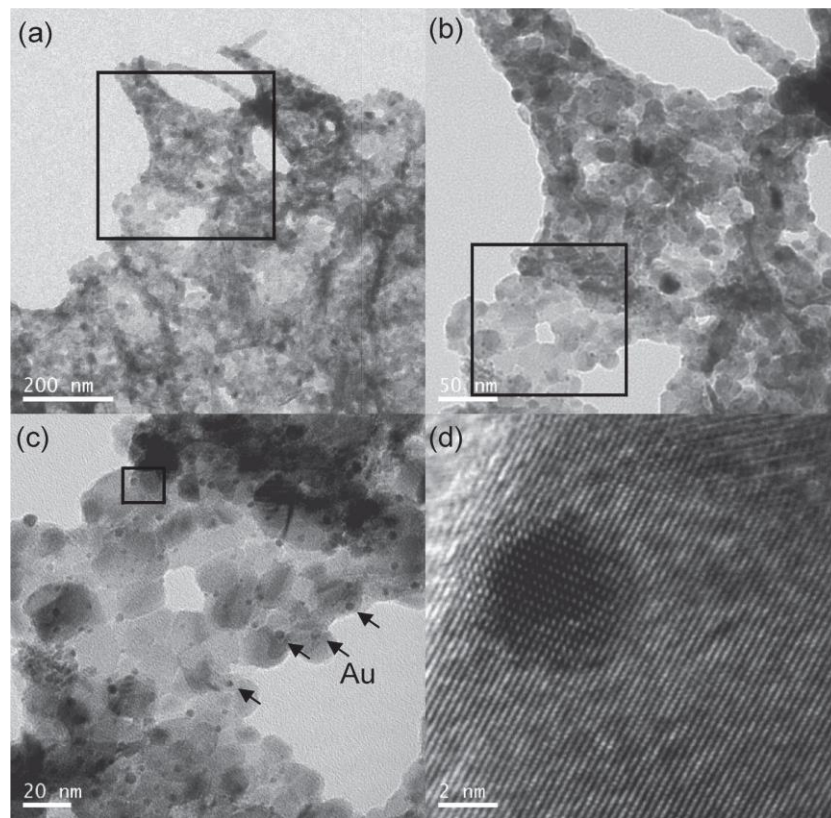
粒徑小於10 nm的金屬粒子分散於一具有多孔網狀結構的複合氧化物擔體中  
藉由調整照光條件  
可在光觸媒上還原出  
奈米級的觸媒顆粒(金、鈀或鉑)

$A_xB_yO_z$

A:銀(Ag)、銅(Cu)、鈀(Pd)、鉑(Pt)

B:鋁(Al)、鈦(Sc)、鉻(Cr)、釔(Y)、鐵(Fe)、銦(In)、鎵(Ga)、鈷(Co)、錳(Mn)、銦(Rh)、鎳(Ni)、鐳(La)、釹(Nd)、釷(Sm)、鎔(Eu)、鈦(Ti)

A離子及B離子的金屬鹽或有機金屬化合物，混合後作為金屬前驅物；將該金屬前驅物溶解於一溶劑中，形成一金屬前驅物溶液；將一燃料加入該金屬前驅物溶液中混合，其中，該燃料選自胺基酸類燃料、草酸或硼烷；蒸發該金屬前驅物溶液中的多餘水分，形成膠體；及加熱該膠體，使該膠體產生自燃反應，以在反應完畢後，得到該氧化物 $A_xB_yO_z$ 粉末，其中將該膠體加熱到使該膠體產生自燃反應的溫度大約150至250°C。



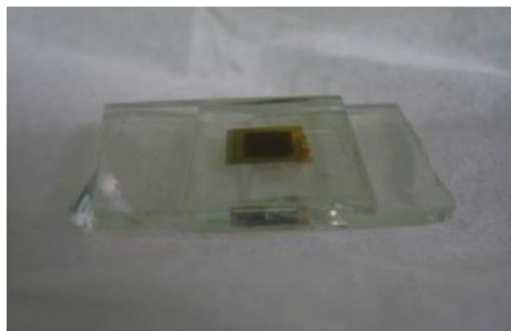
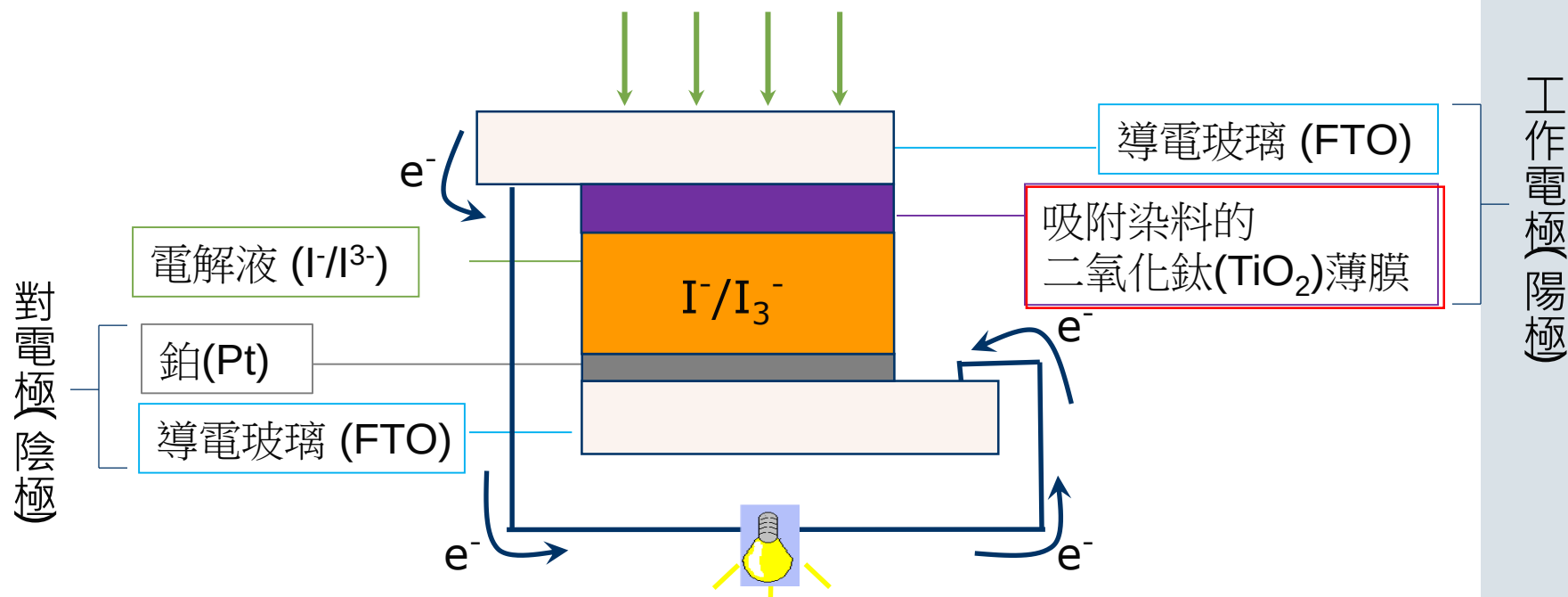
(邱德威、楊重光)、蘇昭瑾



# 回收稀土元素的應用技術

台日資源循環技術交流研討會

## ❖ 染料敏化太陽能電池-染料敏化太陽能電池的結構

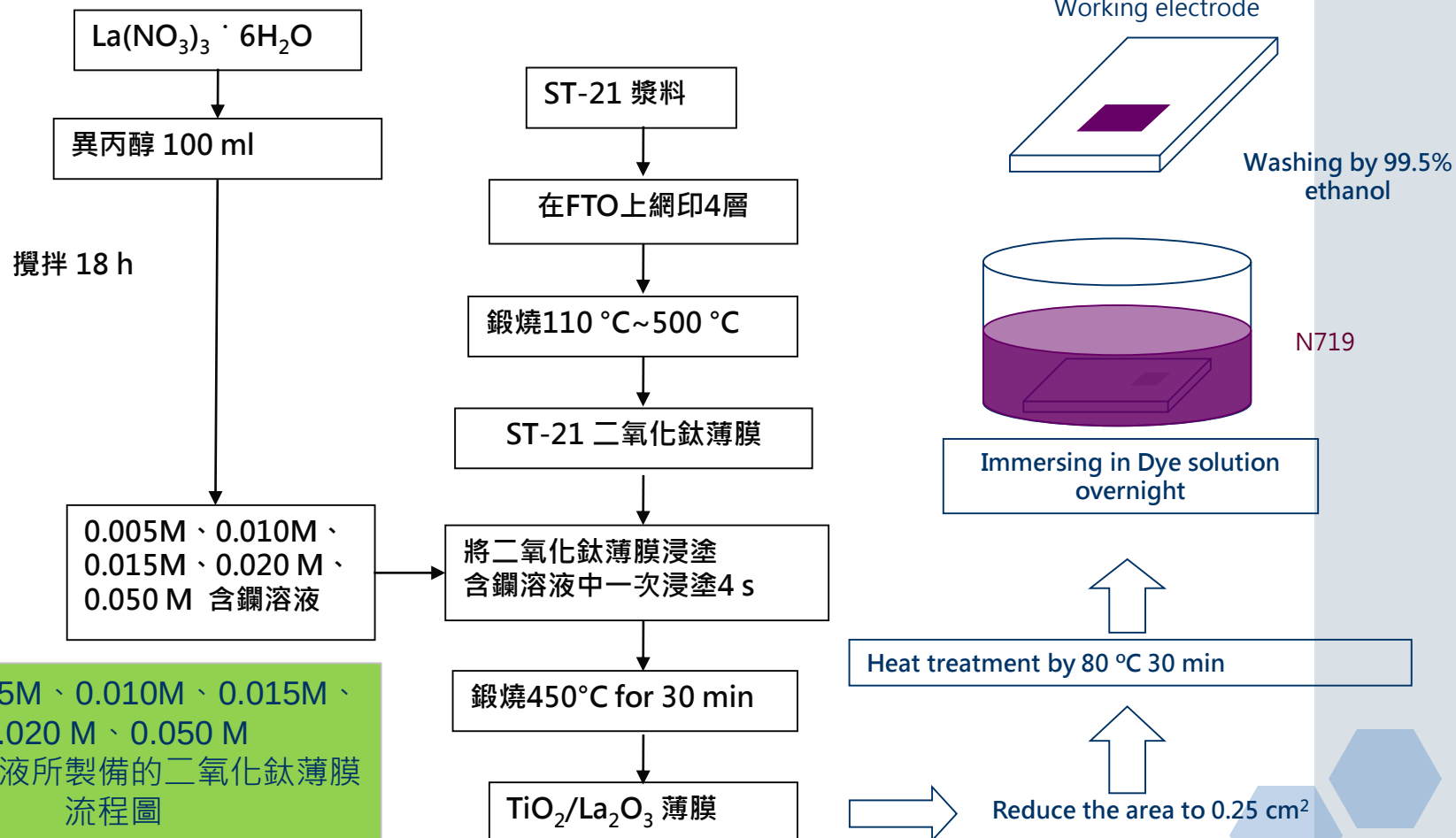




# 含稀土金屬的光陽極製備

台日資源循環技術交流研討會

## ❖ 浸塗法製備含稀土的二氧化鈦薄膜-鐳





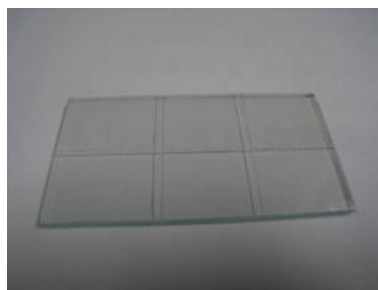
# 陰極的製備

台日資源循環技術交流研討會

## ❖ 陰極(相對電極)製備流程



以超音波震盪槽  
清洗FTO



洗淨後的FTO



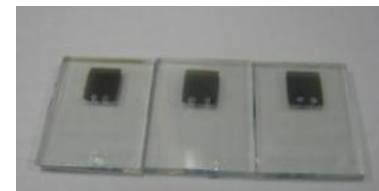
將FTO以賤鍍機  
鍍上Pt



用鑽孔機將Pt  
電極鑽2個孔



以超音波震盪槽  
清洗Pt電極

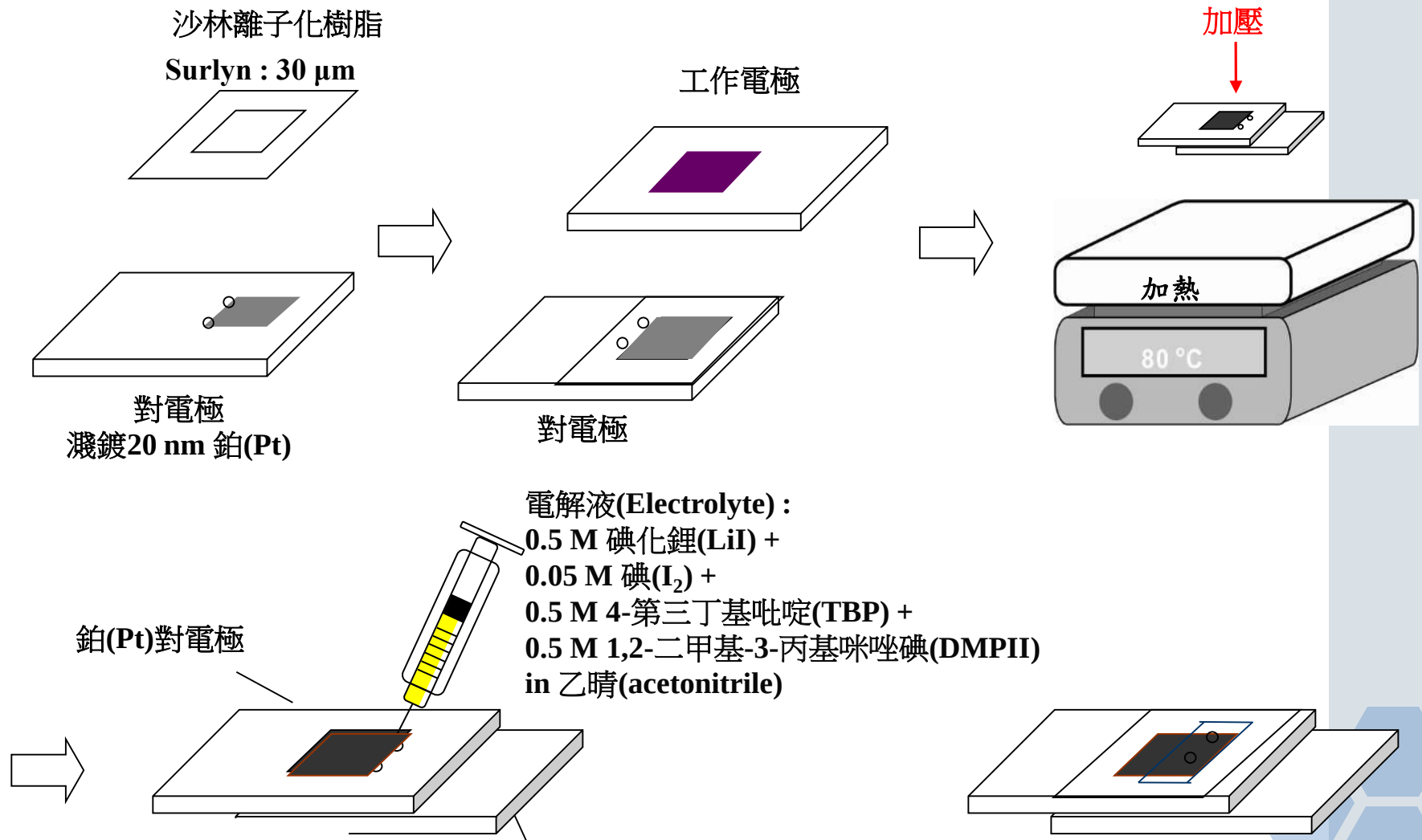


製作完成的  
Pt對電極



# 染料敏化太陽能電池的組裝

台日資源循環技術交流研討會

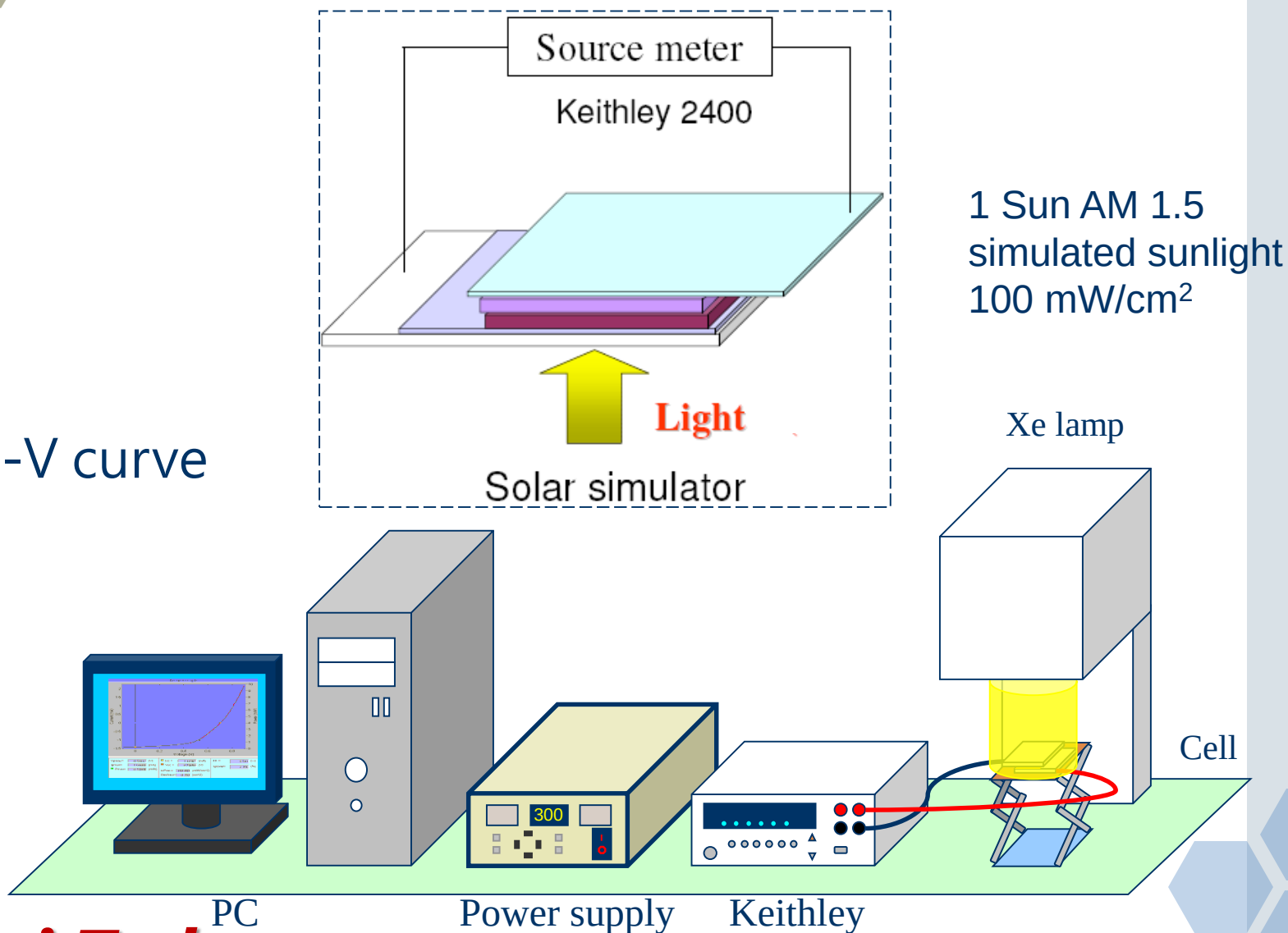




# 染料敏化太陽能電池工作電極性質及光電效率分析

台日資源循環技術交流研討會

## • I-V curve







# 標準品分析

台日資源循環技術交流研討會

## ❖ 含鑷的二氧化鈦薄膜之性質分析-DSSC I-V Curve

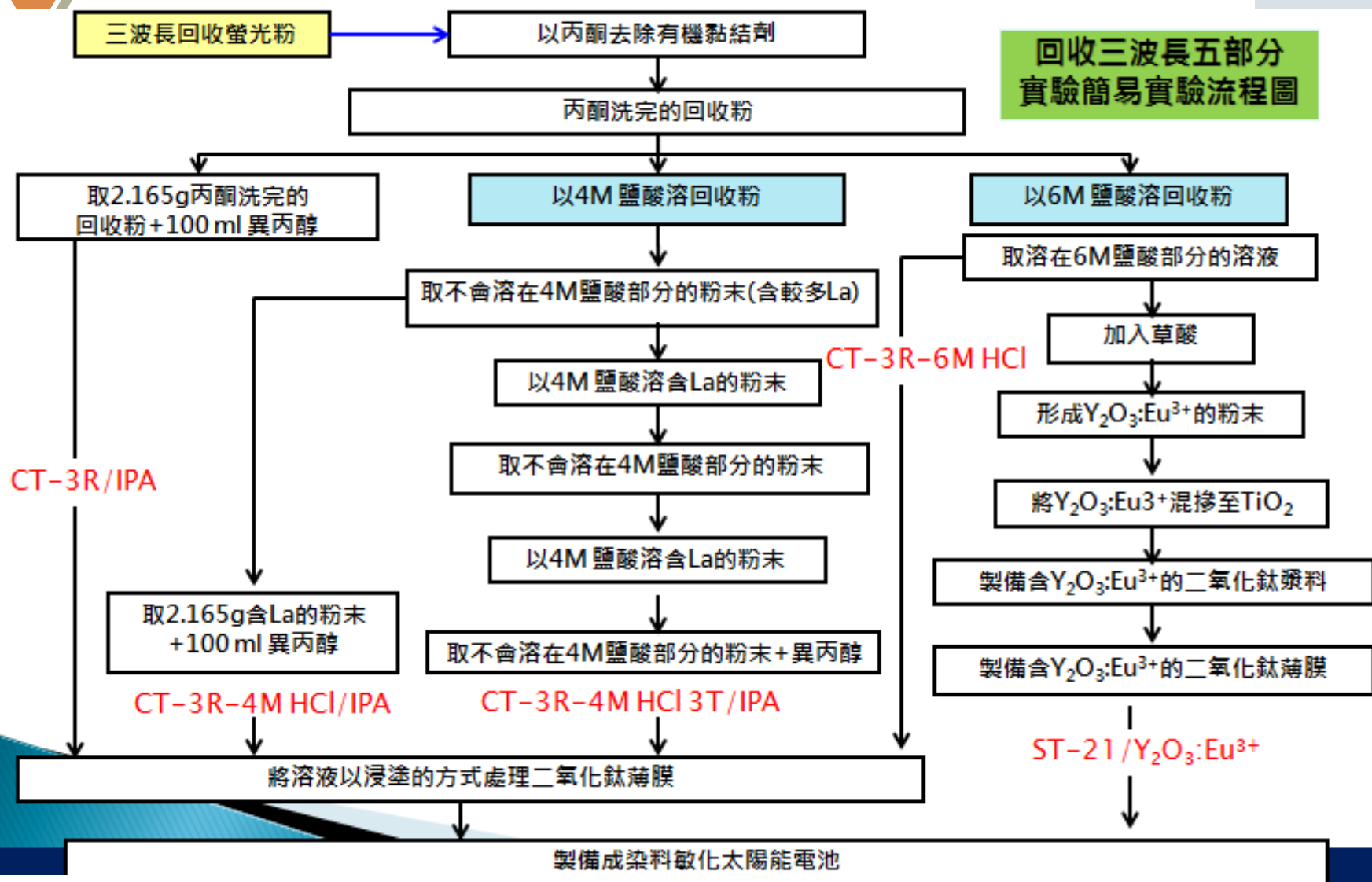
| Sample                                                                            | Thickness (μm) | N719 <sub>ads</sub> (μmole/cm <sup>2</sup> ) | J <sub>sc</sub> (mA/cm <sup>2</sup> ) | V <sub>oc</sub> (V) | FF    | η (%)            |
|-----------------------------------------------------------------------------------|----------------|----------------------------------------------|---------------------------------------|---------------------|-------|------------------|
| ST-21 film                                                                        | 10             | 0.277                                        | 14.7507                               | 0.74                | 0.677 | 7.283<br>(7.098) |
| ST-21 /<br><b>0.005 M</b><br>La(NO <sub>3</sub> ) <sub>3</sub> ·6H <sub>2</sub> O | 9              | 0.264                                        | 14.2933                               | 0.78                | 0.678 | 7.563<br>(7.880) |
| ST-21 /<br><b>0.010 M</b><br>La(NO <sub>3</sub> ) <sub>3</sub> ·6H <sub>2</sub> O | 10             | 0.262                                        | 14.1488                               | 0.76                | 0.679 | 7.298<br>(7.368) |
| ST-21 /<br><b>0.015 M</b><br>La(NO <sub>3</sub> ) <sub>3</sub> ·6H <sub>2</sub> O | 10             | 0.257                                        | 13.0455                               | 0.78                | 0.671 | 6.832            |
| ST-21 /<br><b>0.020 M</b><br>La(NO <sub>3</sub> ) <sub>3</sub> ·6H <sub>2</sub> O | 9              | 0.266                                        | 13.7210                               | 0.78                | 0.669 | 7.084            |
| ST-21 /<br><b>0.050 M</b><br>La(NO <sub>3</sub> ) <sub>3</sub> ·6H <sub>2</sub> O | 10             | 0.264                                        | 11.8347                               | 0.78                | 0.678 | 6.260            |

效率增加範圍為3.8~11%



# 回收品

台日資源循環技術交流研討會

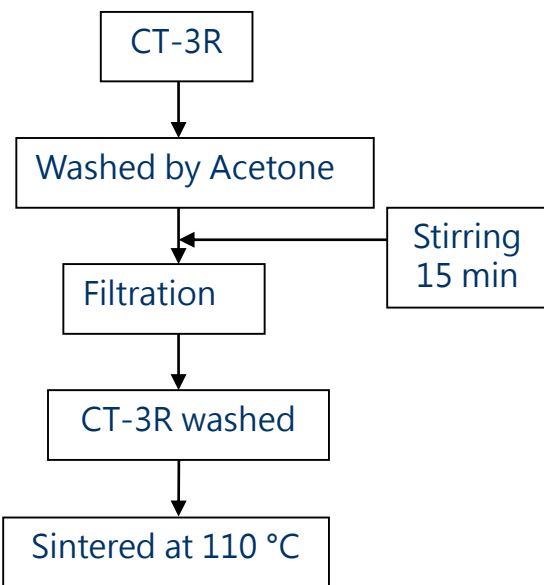




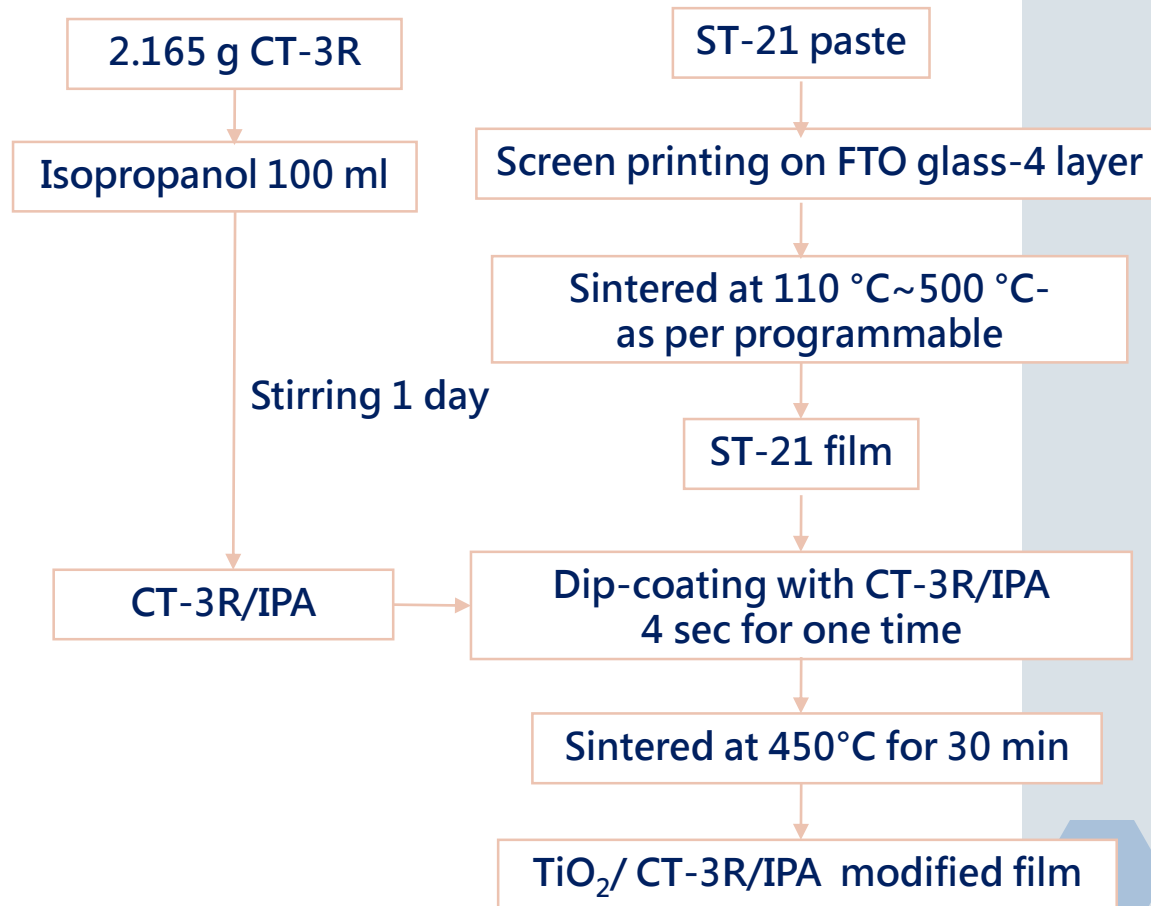
# 回收品

台日資源循環技術交流研討會

❖ 稀有元素 (rare element) : 在地殼中平均含量 $\leq 0.01\%$ 的化學元素



丙酮清洗回收三波長  
螢光粉的實驗流程圖



CT-3R/IPA溶液浸塗ST-21二氧化鈦薄膜實驗流程圖



## ❖ CT-3R-4M HCl-/IPA溶液浸塗ST-21二氧化鈦薄膜實驗

| Sample                                       | Thickness (μm) | Jsc (mA/cm <sup>2</sup> ) | Voc (V) | FF    | η (%) |
|----------------------------------------------|----------------|---------------------------|---------|-------|-------|
| ST-21 film                                   | 12             | 14.8133                   | 0.74    | 0.654 | 7.164 |
|                                              | 11             | 14.5946                   | 0.74    | 0.640 | 6.915 |
| 未稀釋的TiO <sub>2</sub> /<br>CT-3R-4M HCl /IPA  | 11             | 16.1133                   | 0.73    | 0.647 | 7.608 |
|                                              | 12             | 16.0370                   | 0.76    | 0.662 | 8.069 |
| 0.5倍的TiO <sub>2</sub> /<br>CT-3R-4M HCl /IPA | 12             | 15.4593                   | 0.73    | 0.649 | 7.327 |
|                                              | 11             | 15.3264                   | 0.75    | 0.657 | 7.553 |
| 0.2倍的TiO <sub>2</sub> /<br>CT-3R-4M HCl /IPA | 12             | 15.5125                   | 0.73    | 0.643 | 7.280 |
|                                              | 12             | 15.4057                   | 0.74    | 0.644 | 7.338 |
| 0.1倍的TiO <sub>2</sub> /<br>CT-3R-4M HCl /IPA | 11             | 15.0533                   | 0.75    | 0.644 | 7.275 |
|                                              | 10             | 14.4012                   | 0.73    | 0.664 | 7.080 |

- 未稀釋的CT-3R-4M HCl/IPA二氧化鈦薄膜光電換效率最高可達8.069 %。  
其效率增益約為16.6 %。





## 結論

台日資源循環技術交流研討會

- 國外稀土資源資源化以傳統酸萃方法加以有機分子溶劑萃取使得稀土資源資源化技術更加多樣化。
- 針對回收螢光粉改質染料敏化太陽能電池(DSSC)光陽極，以及提升DSSC光電轉換效率進行深入的研究。由回收三波長螢光粉酸萃處理後的含鑷萃餘物改質DSSC光陽極，其光電轉換效率最高可達8.069 %，效率增益最高可提升至16.6 %。
- 本研究成功地將回收螢光粉的主要稀土元素創新應用於染料敏化太陽能電池，並且成為稀土資源資源化與再利用技術的示範佳作。





# 鈷元素回收技術探討

台日資源循環技術交流研討會

| 技術名稱              | 最終產品                                                 | 回收率   | 國家 | 回收物品                 | 優點                                 | 缺點                     | 文獻 |
|-------------------|------------------------------------------------------|-------|----|----------------------|------------------------------------|------------------------|----|
| 酸浸法<br>沉澱法        | $\text{CoC}_2\text{O}_4$<br>$\text{Li}_2\text{CO}_3$ | 94.7% | 中國 | 鋰電池                  | 1.萃取效果好<br>2.同時回收Co和Li             | Li的回收率較低               | 1  |
| 萃取法               | $\text{Co}^{2+}$                                     | 97%   | 印度 | 植物<br>複合材料<br>合金     | 1.能有效分離出Co<br>2.操作簡單<br>3.能應用於實際樣本 | Cyanex 272萃取量低<br>不利量化 | 2  |
| 酸浸法<br>萃取法        | $\text{Co}^{2+}$                                     | 95%   | 印度 | Co-Mn催化劑<br>Co-Mo催化劑 | 1.萃取效果好<br>2.純度高<br>3.設備需求低        | 須經由多次的萃取<br>較繁瑣        | 3  |
| 酸浸法<br>萃取法<br>沉澱法 | $\text{CoC}_2\text{O}_4$                             | 93%   | 中國 | 鋰電池                  | 1.純度高<br>2.操作簡單                    | 需較多儀器設備<br>步驟繁瑣        | 4  |
| 酸浸法               | $\text{Co}^{2+}$                                     | 84.5% | 中國 | 紅土礦                  | 1.步驟簡單                             | 氟化氫銨有毒<br>無法單一萃取出Co    | 5  |



## 參考文獻

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- ❖ 2. Research Journal of Chemical Sciences Vo1. 2(1), 75-82, Jan. (2012)
- ❖ 3. Indian Journal of Chemistry Vol. 42A, Dec. 2003, pp.2954-2958
- ❖ 4. Hydrometallurgy 108 (2011) 80-86
- ❖ 5. Trans. Nonferrous Met. Soc. China 20(2010) 1517-1520

- 未稀釋的CT-3R-4M HCl/IPA二氧化鈦薄膜光電換效率最高可達8.069 %。其效率增加約為16.6 %。

# Recovery of Co And Li from Spent Lithium-ion Batteries by Combination Method of Acid Leaching and Chemical Precipitation

台日資源循環技術交流研討會

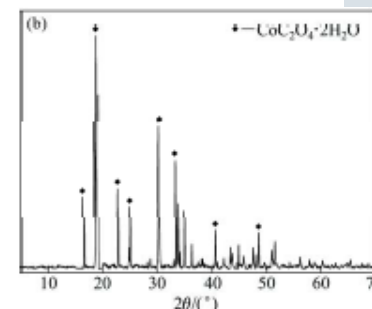
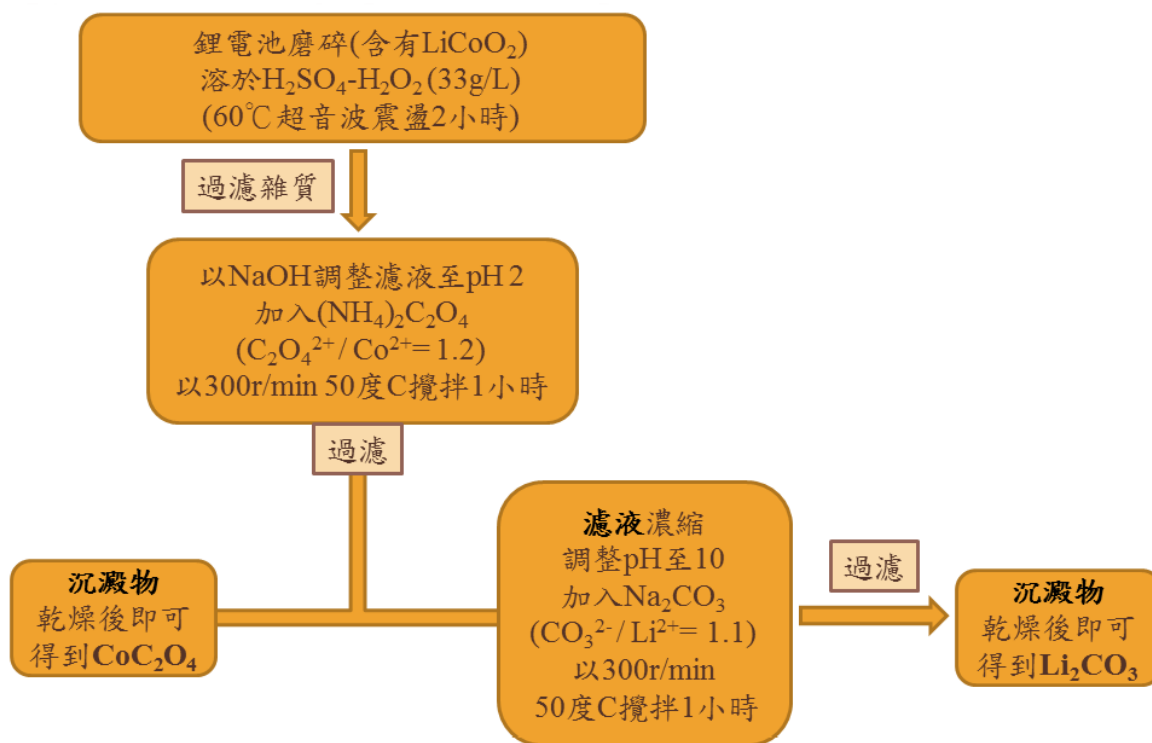


Fig. 4 Photo (a) and XRD pattern (b) of products of CoC<sub>2</sub>O<sub>4</sub>·2H<sub>2</sub>O obtained by sedimentation

Table 1 Composition analysis of CoC<sub>2</sub>O<sub>4</sub> samples (mass fraction, %)

| Fe    | Co    | Ni    | Mn    | As    | Cu    | Ca    | Na    |
|-------|-------|-------|-------|-------|-------|-------|-------|
| 0.001 | 32.69 | 0.015 | 0.003 | 0.002 | 0.004 | 0.001 | 0.006 |

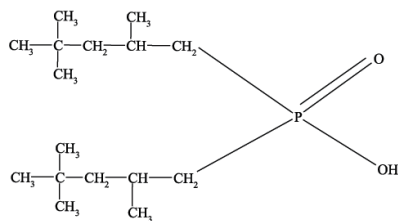
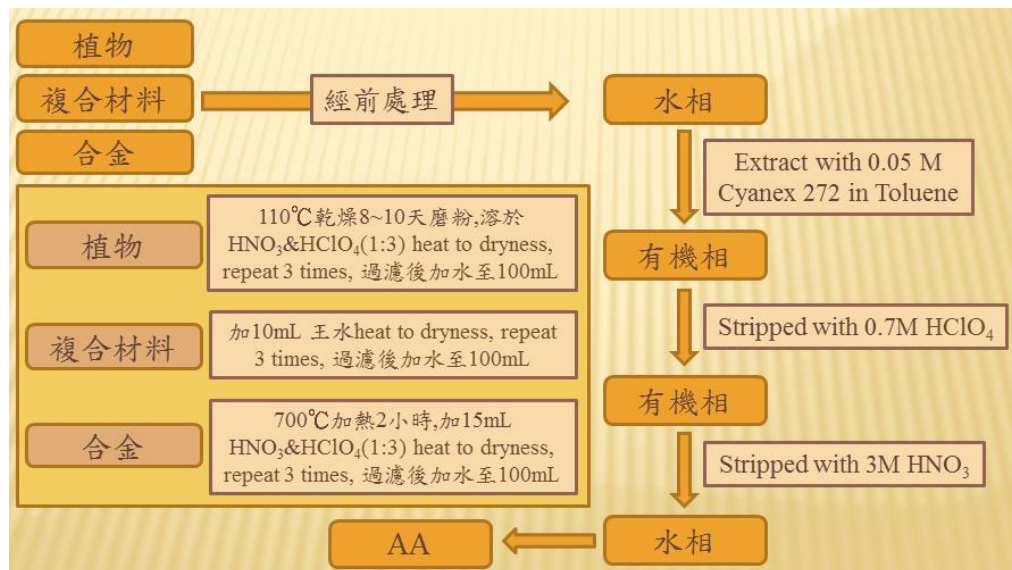
結論：

1. 操作簡單
2. Co的回收率可達94.7%
3. 回收物純度高

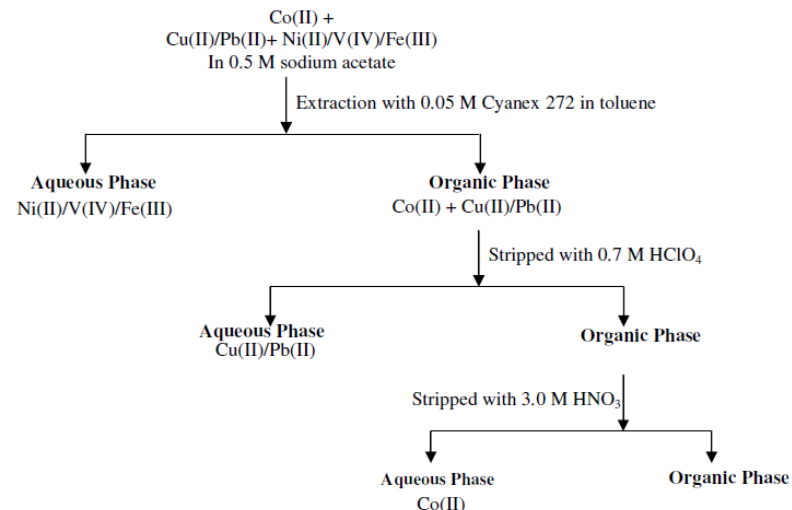


# Liquid-Liquid Extraction and Separation of Cobalt(II) from Sodium Acetate media using Cyanex 272

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CYANEX 272



結論：

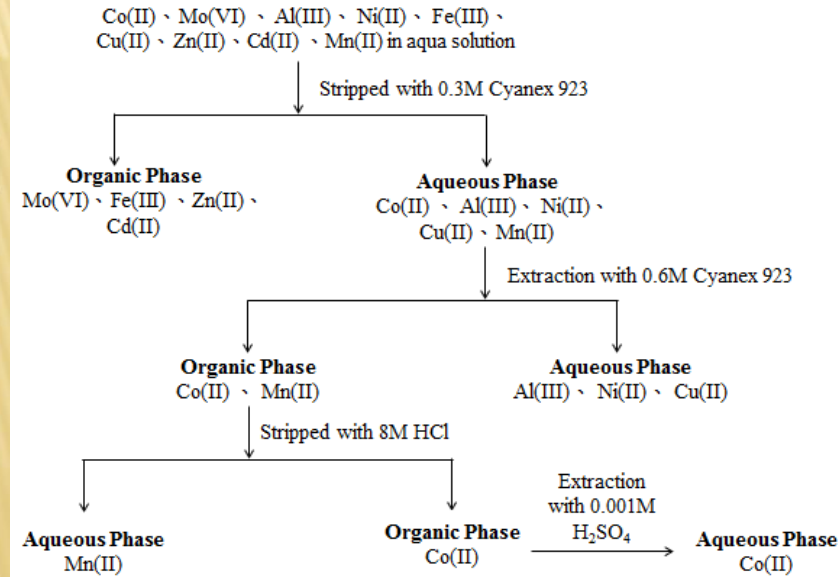
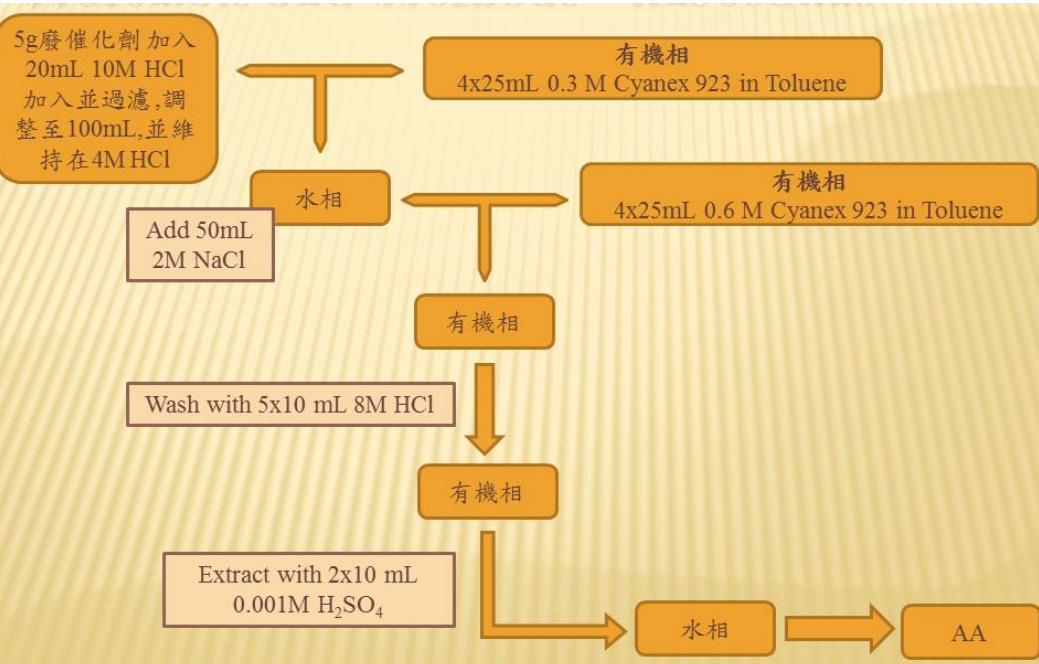
- 1.操作簡單
- 2.能從不同金屬有效分離
- 3.Co回收率可達97%以上

2. Research Journal of Chemical Sciences Vo1. 2(1), 75-82, Jan. (2012)

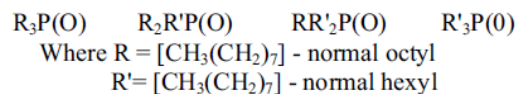


# Recovery of cobalt from secondary sector using extraction in Cyanex 923

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CYANEX 923 extractant is a mixture of four trialkyl-phosphine oxides as follows:



結論：

- 1.需經多次萃取
- 2.回收率可達95%以上
- 3.純度可達99%以上

3. Indian Journal of Chemistry Vol. 42A, Dec. 2003, pp.2954-2958)

國立臺北科技大學稀土與稀有元素技術研發中心



# Process for the Recovery of Cobalt Oxalate from Spent Lithium-ion Batteries – (1)

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取出鋰電池的陰、陽極，磨碎，150~200°C，2~3小時

再磨成粉，溶於5 wt% NaOH，室溫4小時

remove Al(III)

filter

Residue 於700~800°C，4~5小時

remove carbon and binder

Residue

Co(II)、Ni(II)、Fe(III)、Cu(II)、Li(I)、Mn(II)

Residue leaching with 4M H<sub>2</sub>SO<sub>4</sub> /10% H<sub>2</sub>O<sub>2</sub> 85°C, 2hr

浸出液

Co(II)、Ni(II)、Fe(III)、Cu(II)、Li(I)、Mn(II)

filter

Add 10% NaOH to pH 3.0, 95°C, 2hr

filter

**Precipitate**  
Sodium jorosite  
remove Fe(III)

Adjust to pH 4.0, add 10% (NH<sub>4</sub>)<sub>2</sub>S<sub>2</sub>O<sub>8</sub>, 70°C, 2hr

filter

**Precipitate**  
MnO<sub>2</sub>  
remove Mn(II)

Add 10% NaOH to pH 5.5

filter

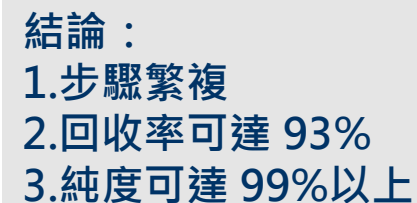
**Precipitate**  
Cu(OH)<sub>2</sub>  
remove Cu(II)

Co(II)、Ni(II)、Li(I) in Aqua solution

4. Hydrometallurgy 108 (2011) 80–86



台日資源循環技術交流研討會

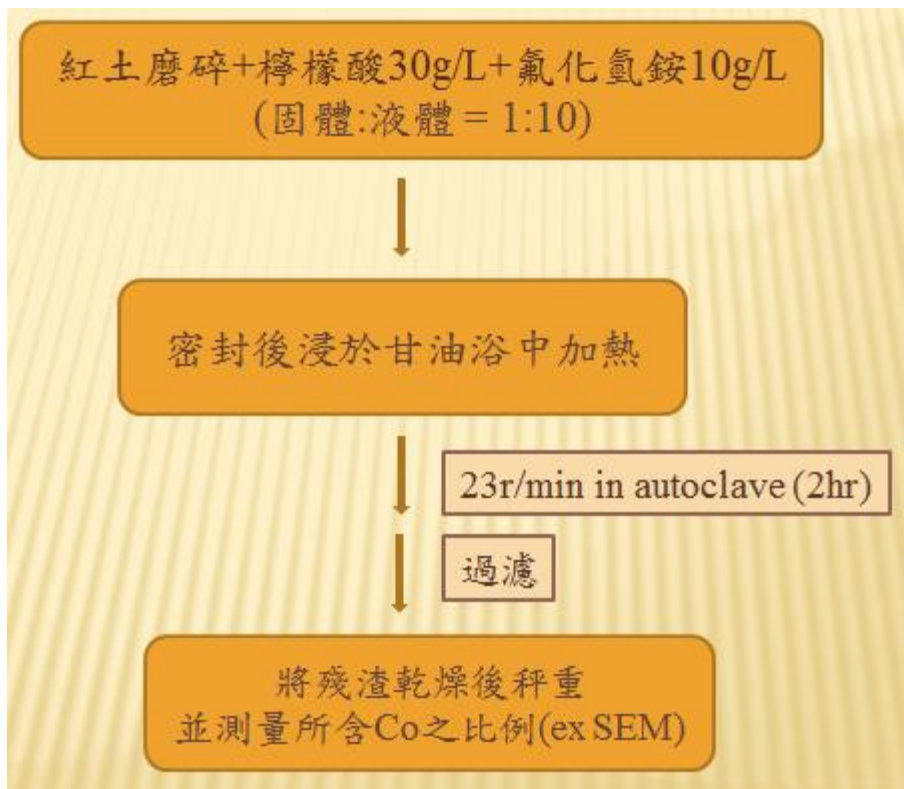


4. Hydrometallurgy 108 (2011) 80–86



# Extraction of Cobalt from laterite ores by Citric Acid in presence of Ammonium Bifluoride

台日資源循環技術交流研討會



**Table 1** Main chemical compositions of laterite (mass fraction, %)

| Co   | Ni   | MgO  | Al <sub>2</sub> O <sub>3</sub> | MnO <sub>2</sub> | SiO <sub>2</sub> | Total iron grade |
|------|------|------|--------------------------------|------------------|------------------|------------------|
| 0.13 | 1.03 | 0.98 | 9.72                           | 1.25             | 4.25             | 43.95            |

**Table 2** Main chemical compositions of leached residue (mass fraction, %)

| Co    | Ni   | MgO  | Al <sub>2</sub> O <sub>3</sub> | MnO <sub>2</sub> | SiO <sub>2</sub> | Total iron grade |
|-------|------|------|--------------------------------|------------------|------------------|------------------|
| 0.025 | 0.87 | 0.76 | 9.36                           | 0.48             | 2.23             | 43.18            |

結論：

1.步驟簡單

2.回收率84.5%

5. Trans. Nonferrous Met. Soc. China 20(2010) 1517-1520



# Thank You!

臺北科技大學



**Taipei Tech**